



SOICT



Multimodal Machine Learning

Lecture 3.2: Multimodal Coordination and Fission

Đàm Quang Tuấn

Objectives of today's class

- Representation coordination
 - Coordination functions
 - Kernel similarity functions
 - Canonical correlation analysis
 - Contrastive learning
 - Information, entropy and mutual information
- Representation fission
 - Factorized multimodal representations
 - Clustering and fine-grained fission

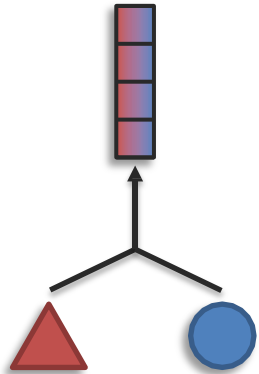
Multimodal Representation

Challenge 1: Representation

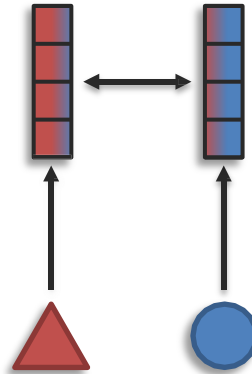
Definition: Learning representations that reflect cross-modal interactions between individual elements, across different modalities

Sub-challenges:

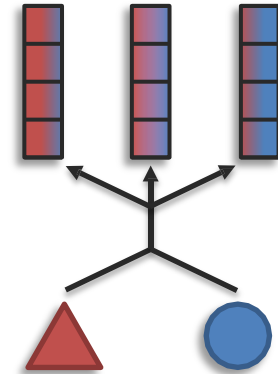
Fusion



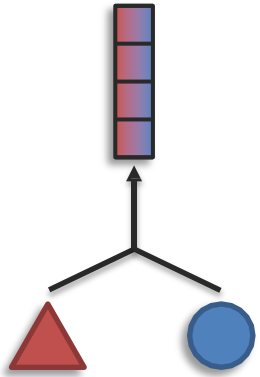
Coordination



Fission

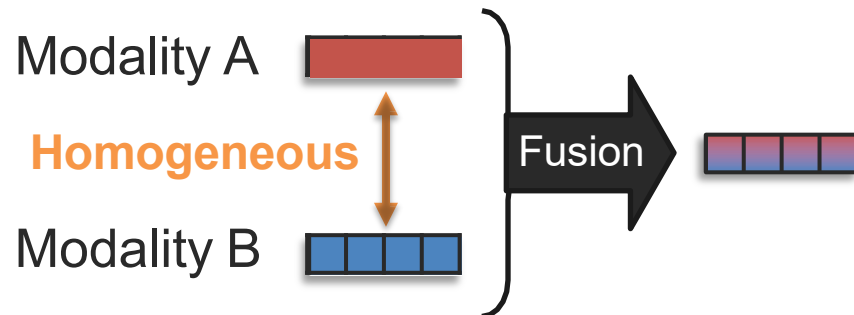


Sub-Challenge 1a: Representation Fusion

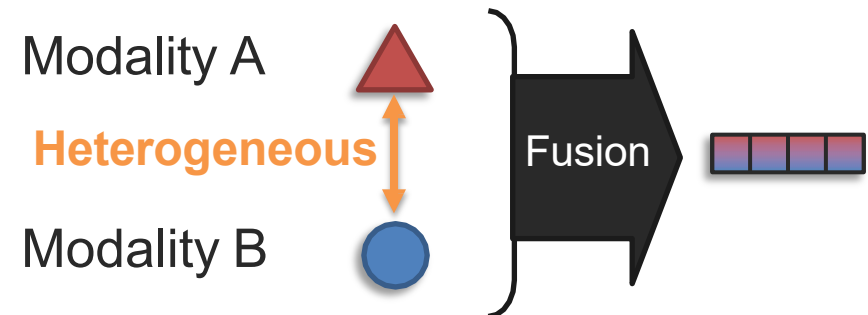


Definition: Learn a joint representation that models cross-modal interactions between individual elements of different modalities

Basic fusion:



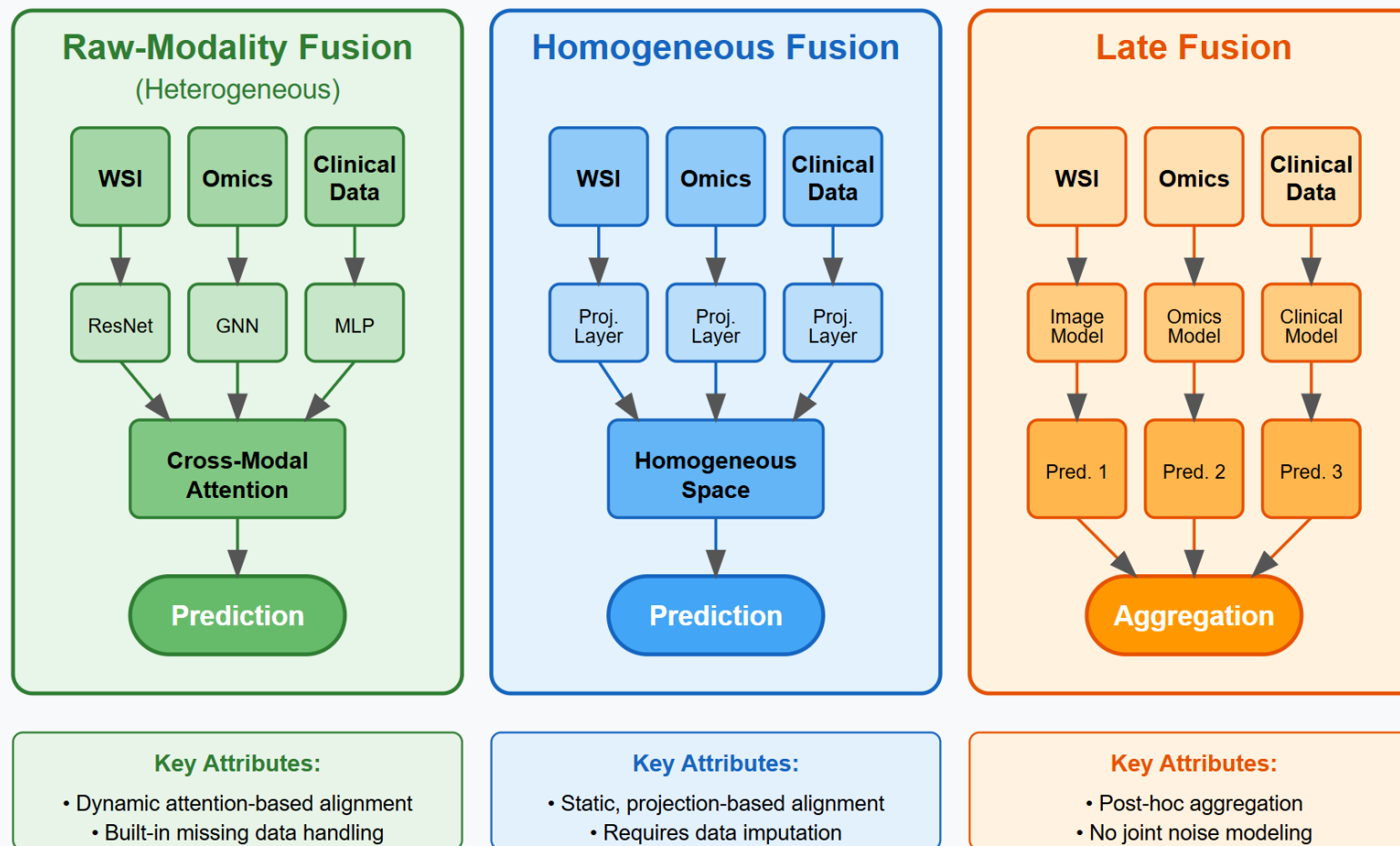
Raw-modality fusion:



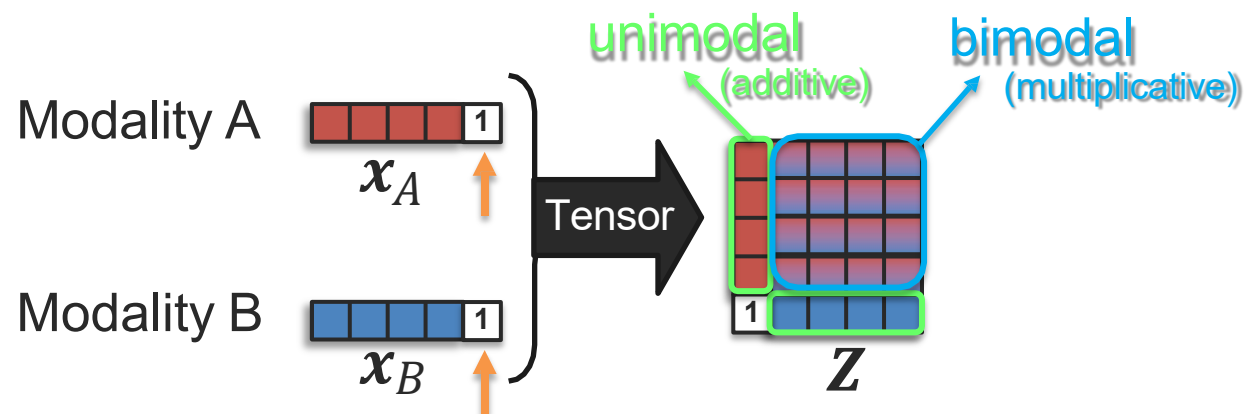
Sub-Challenge 1a: Representation Fusion

Raw-Modality Fusion vs. Other Approaches

Comparing Different Multimodal Fusion Strategies

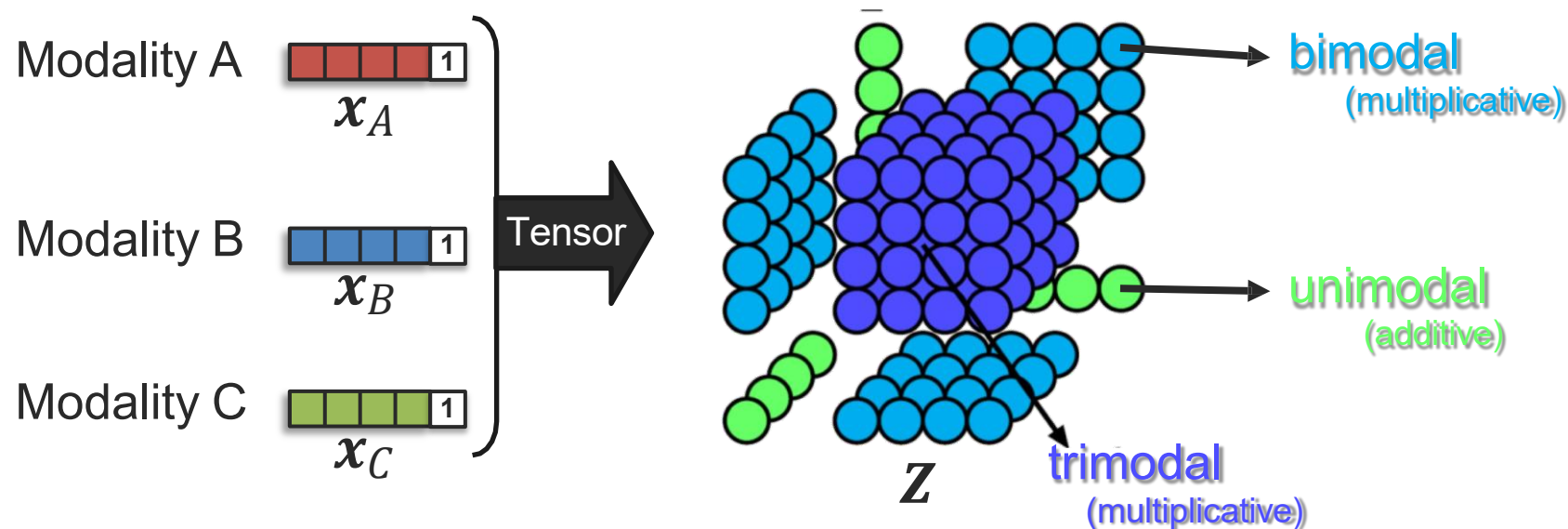


Tensor Fusion



Tensor Fusion (bimodal):

$$Z = w([x_A \ 1]^T / [x_B \ 1])$$



... but the weight matrix may end up quite large!

Low-rank Fusion

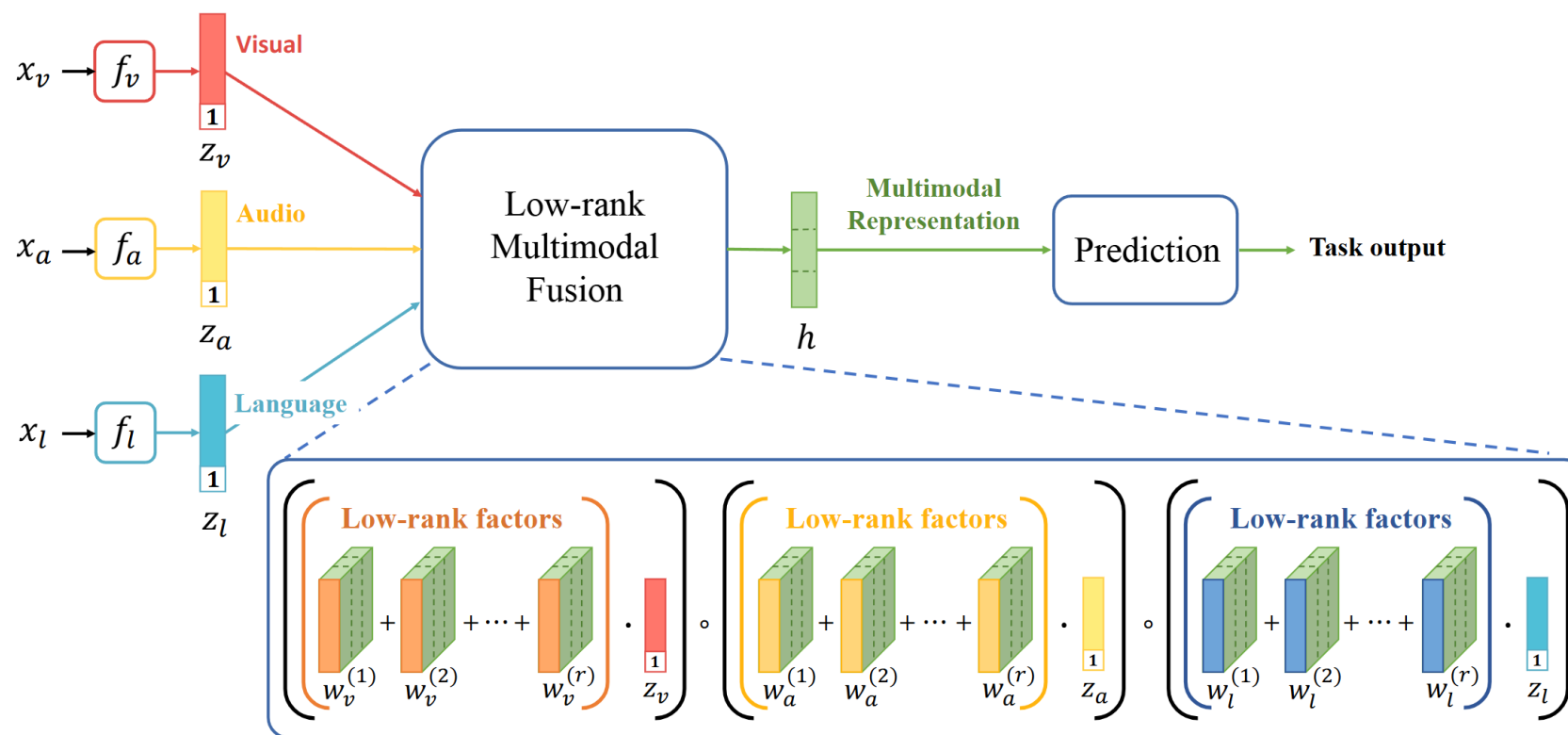
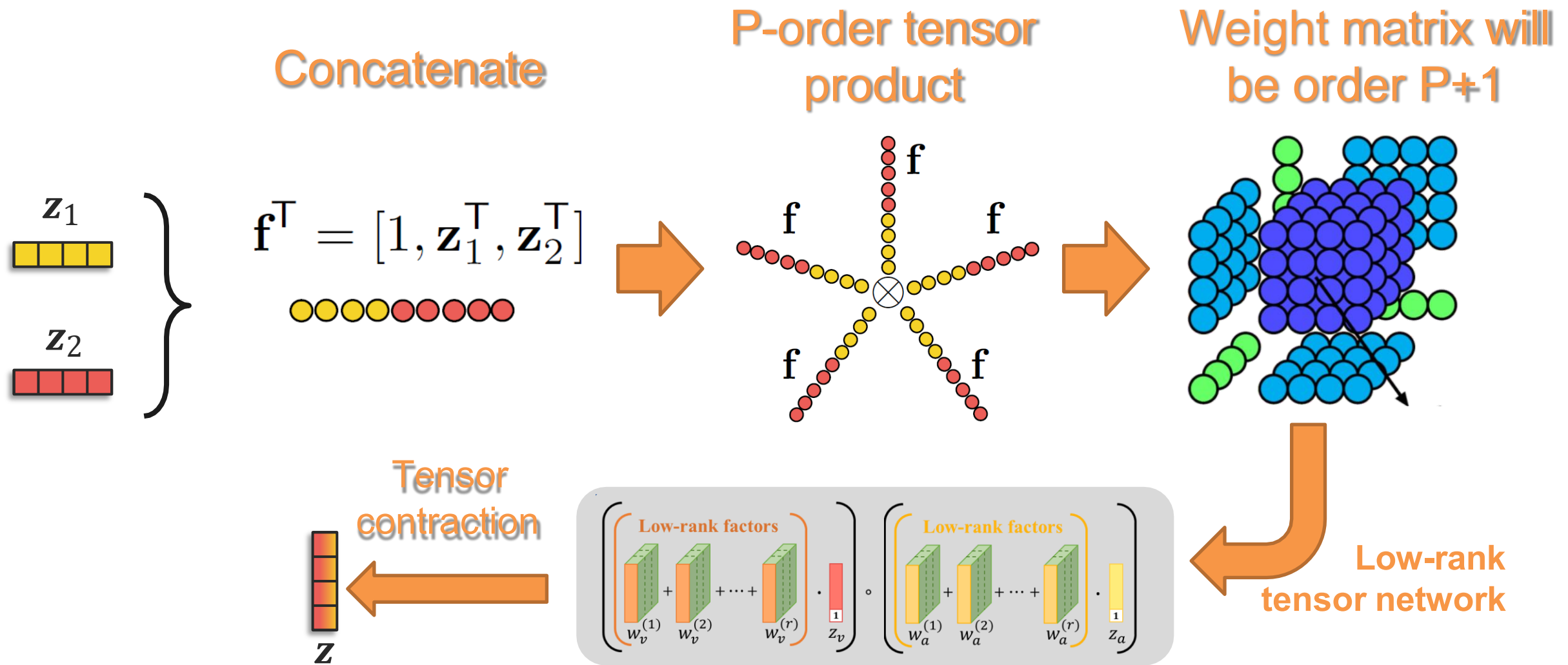
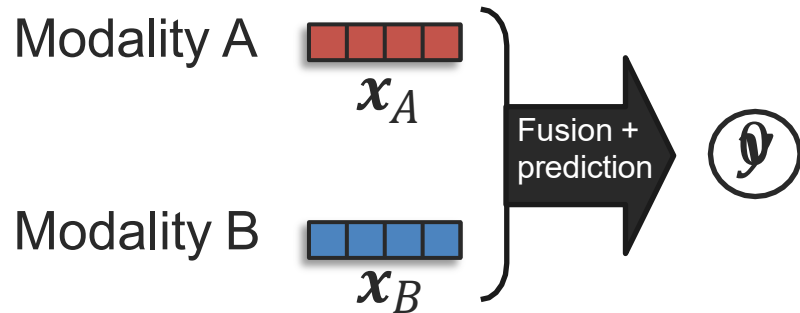


Figure 1: Overview of our Low-rank Multimodal Fusion model structure: LMF first obtains the unimodal representation z_a, z_v, z_l by passing the unimodal inputs x_a, x_v, x_l into three sub-embedding networks f_v, f_a, f_l respectively. LMF produces the multimodal output representation by performing low-rank multimodal fusion with modality-specific factors. The multimodal representation can be then used for generating prediction tasks.

High-Order Polynomial Fusion



Measuring Non-Additive Interactions



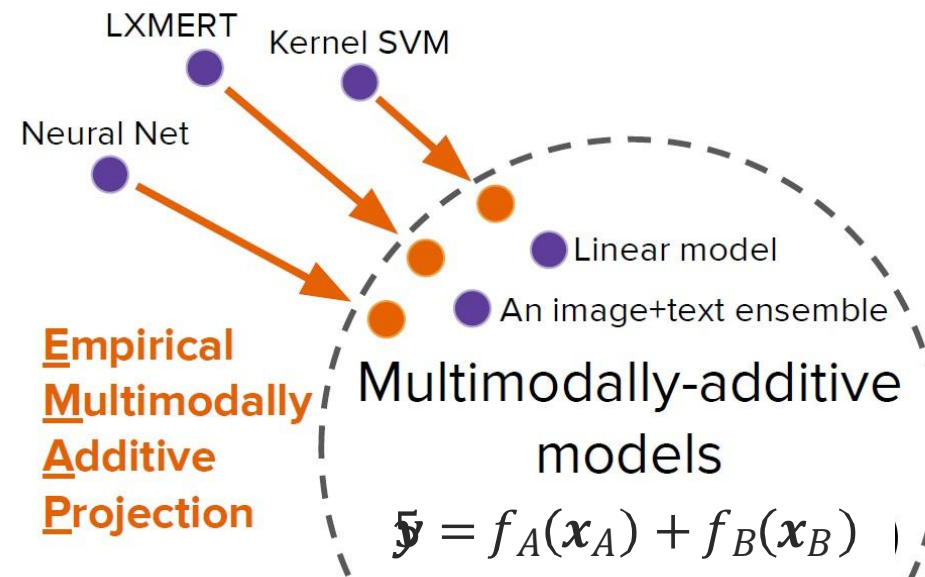
Nonlinear fusion:

$$y = f(x_A, x_B)$$

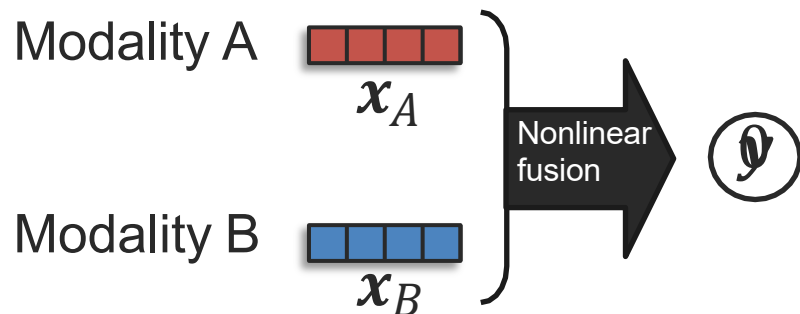
Projection?

Additive fusion:

$$y = f_A(x_A) + f_B(x_B)$$



Measuring Non-Additive Interactions



Nonlinear fusion:

$$y = f(x_A, x_B)$$

Additive fusion:

$$y' = f_A(x_A) + f_B(x_B)$$

Projection?

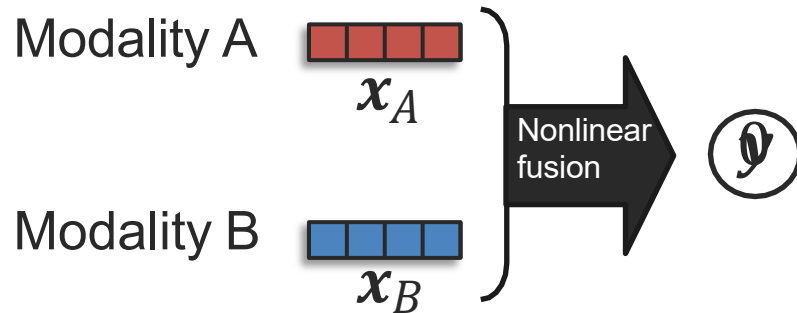
Projection from nonlinear to additive (using EMAP):

$$\hat{y}(x_A, x_B) = \underbrace{E_{x_B} [f(x_A, x_B)]}_{f_A(x_A)} + \underbrace{E_{x_A} [f(x_A, x_B)]}_{f_B(x_B)}$$

Modality A Modality B

Additive fusion
(approximation)

Measuring Non-Additive Interactions



Nonlinear fusion:

$$\phi = f(x_A, x_B)$$

EMAP
projection

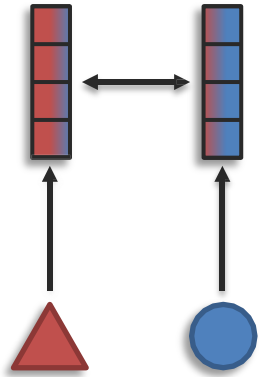
Additive fusion:

$$\phi^@ = f_A(x_A) + f_B(x_B) + \mu$$

		I-INT	I-SEM	I-CTX	T-VIS	R-POP	T-ST1	T-ST2	
Nonlinear	Neural Network	90.4	69.2	78.5	51.1	63.5	71.1	79.9	
Polynomial	Polykernel SVM	91.3	74.4	81.5	50.8	–	72.1	80.9	
Nonlinear	FT LXMERT	83.0	68.5	76.3	53.0	63.0	66.4	78.6	
Nonlinear	$\hookrightarrow$ + Linear Logits	89.9	73.0	80.7	53.4	64.1	75.5	80.3	
Additive	Linear Model	90.4	72.8	80.9	51.3	63.7	75.6	76.1	
	Best Model	91.3	74.4	81.5	53.4	64.2	75.5	80.9	
Additive	$\hookrightarrow$ + EMAP	91.1	74.2	81.3	51.0	64.1	75.9	80.7	Always a good baseline! Differences are small!!!!

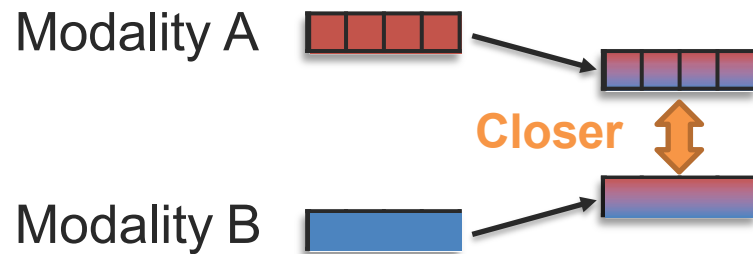
Representation Coordination

Sub-Challenge 1b: Representation Coordination

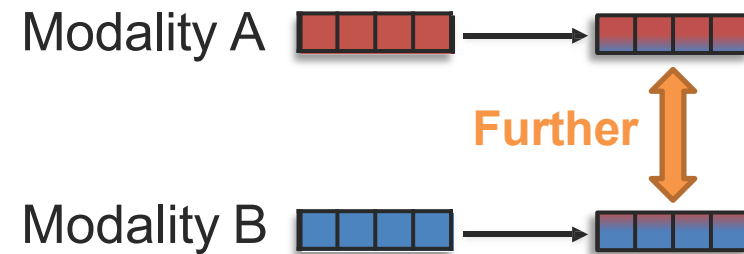


Definition: Learn multimodally-contextualized representations that are coordinated through their cross-modal interactions

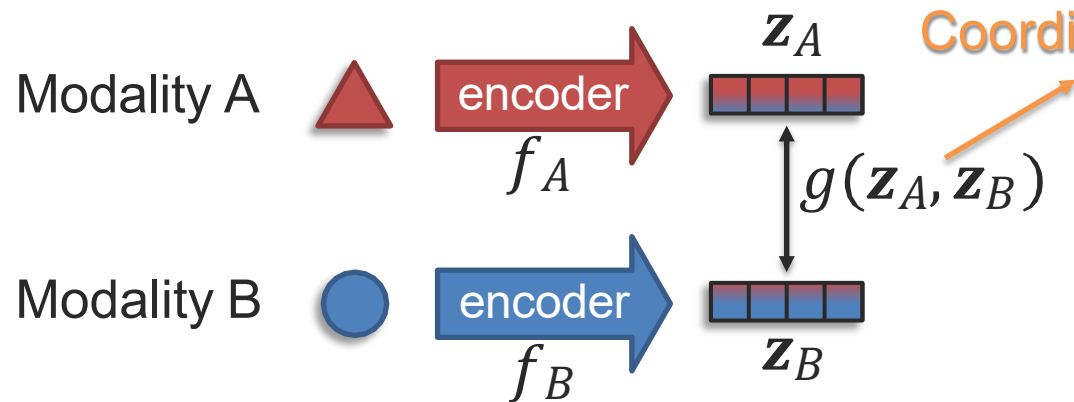
Strong Coordination:



Partial Coordination:



Coordination Function



Learning with coordination function:

$$\mathcal{L} = g(f_A(\triangle), f_B(\circ))$$

with model parameters θ_g , θ_{f_A} and θ_{f_B}

➡ Requires paired data

Examples of coordination function:

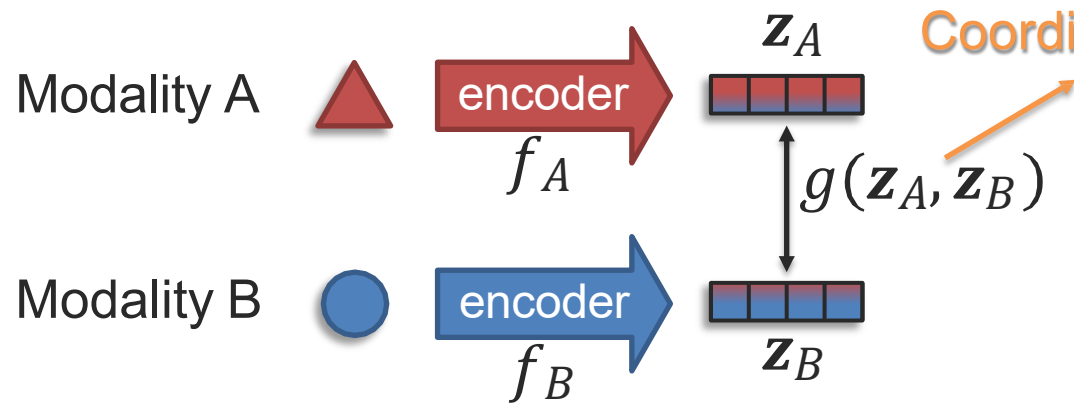
1 Cosine similarity:

$$g(\mathbf{z}_A, \mathbf{z}_B) = \frac{\mathbf{z}_A \cdot \mathbf{z}_B}{\|\mathbf{z}_A\| \|\mathbf{z}_B\|}$$

Strong coordination!

➡ For normalized inputs (e.g., $\mathbf{z}_A - \mathbf{z}_A$), equivalent to *Pearson correlation coefficient*

Coordination Function



Learning with coordination function:

$$\mathcal{L} = g(f_A(\triangle), f_B(\circ))$$

with model parameters θ_g , θ_{f_A} and θ_{f_B}

Examples of coordination function:

2 Kernel similarity functions:

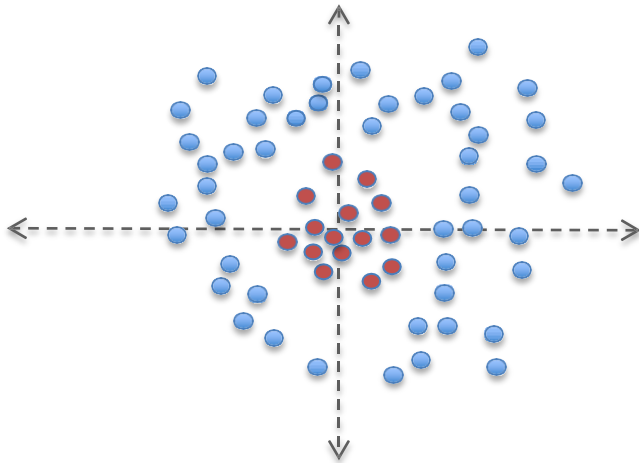
$$g(\mathbf{z}_A, \mathbf{z}_B) = k(\mathbf{z}_A, \mathbf{z}_B) \left\{ \begin{array}{l} \bullet \text{ Linear} \\ \bullet \text{ Polynomial} \\ \bullet \text{ Exponential} \\ \bullet \text{ RBF} \end{array} \right.$$

All these examples bring relatively strong coordination between modalities

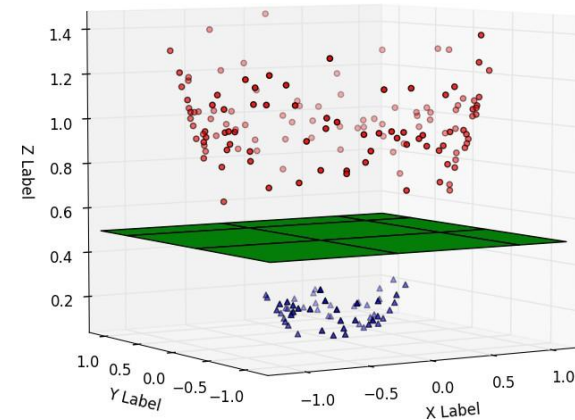
Kernel Function

A kernel function: Acts as a similarity metric between data points

$$K(\mathbf{x}_i, \mathbf{x}_j) = \phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j) = \langle \phi(\mathbf{x}_i), \phi(\mathbf{x}_j) \rangle \quad \Rightarrow \phi(\mathbf{x}) \text{ can be high-dimensional space!}$$



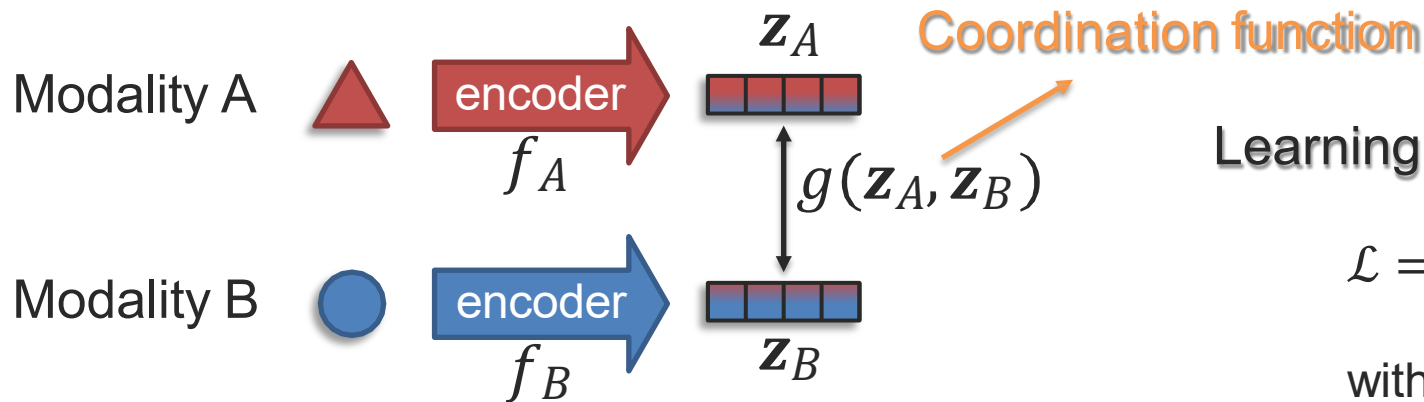
Not linearly separable in $\mathbf{x}$ space



Same data, but now linearly separable in $\phi(\mathbf{x})$ space

➡ Radial Basis Function (RBF) Kernel : $K(\mathbf{x}_i, \mathbf{x}_j) = \exp - \frac{1}{2\sigma^2} \|\mathbf{x}_i - \mathbf{x}_j\|^2$

Coordination Function



Learning with coordination function:

$$\mathcal{L} = g(f_A(\triangle), f_B(\circ))$$

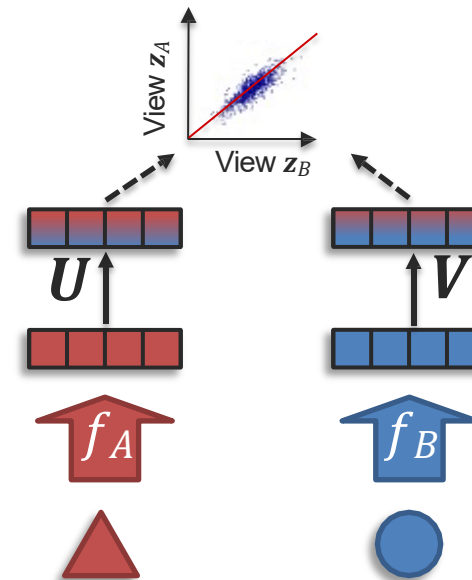
with model parameters θ_g , θ_{f_A} and θ_{f_B}

Examples of coordination function:

③ Canonical Correlation Analysis (CCA):

$$\underset{V, U, f_A, f_B}{\operatorname{argmax}} \operatorname{corr}(\mathbf{z}_A, \mathbf{z}_B)$$

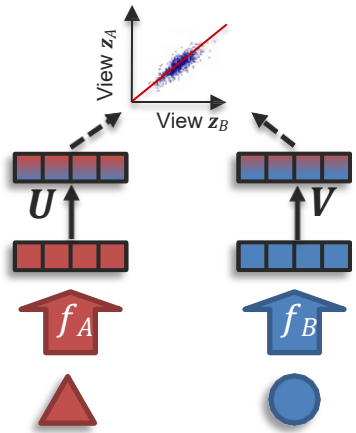
→ CCA includes multiple projections, all orthogonal with each others



Correlated Projection

- 1 Learn two linear projections, one for each view, that are maximally correlated:

$$(\mathbf{u}^*, \mathbf{v}^*) = \operatorname{argmax}_{\mathbf{u}, \mathbf{v}} \operatorname{corr}(\mathbf{u}^T \mathbf{X}, \mathbf{v}^T \mathbf{Y})$$



Two views X, Y where same instances have the same color

➡ Remember that X and Y consist of paired data

Deep Canonically Correlated Autoencoders (DCCAE)

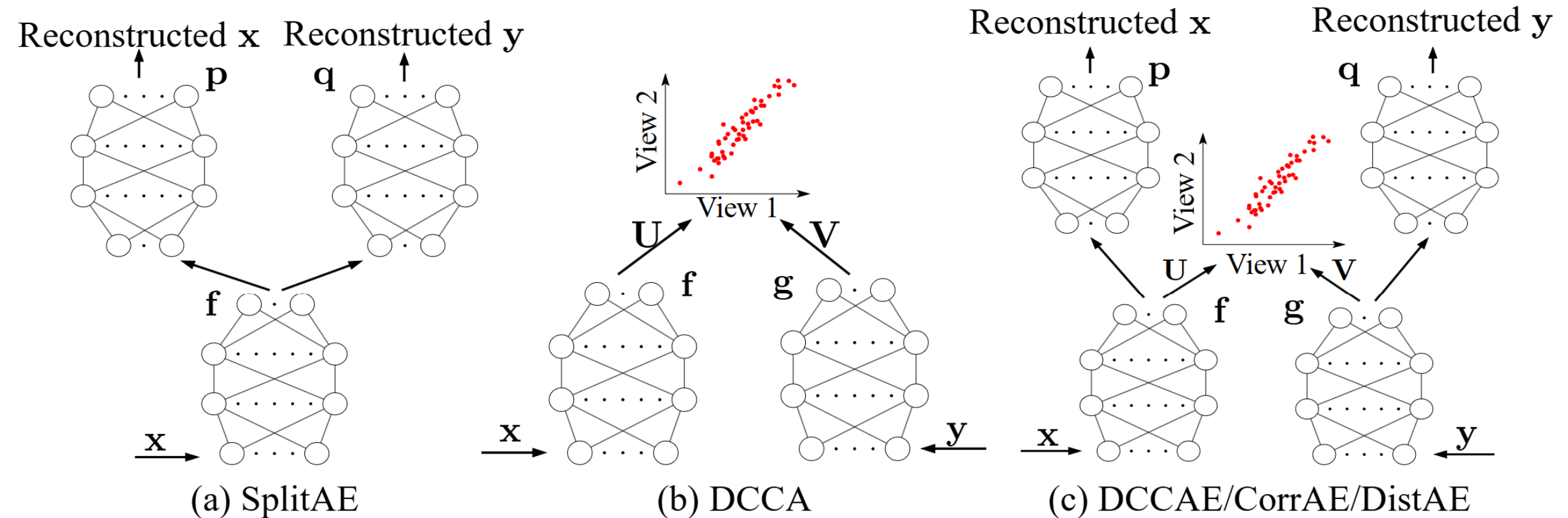
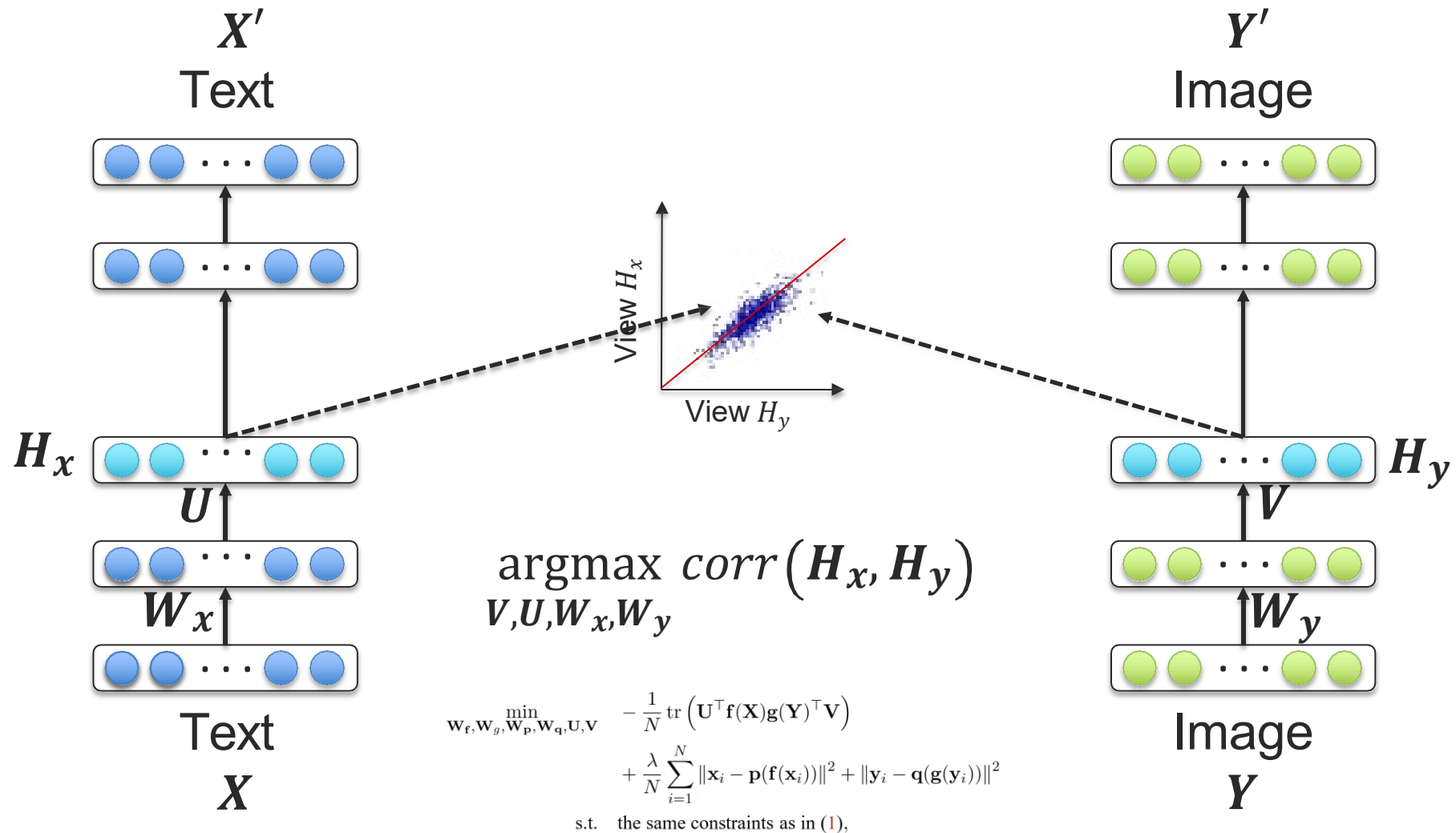


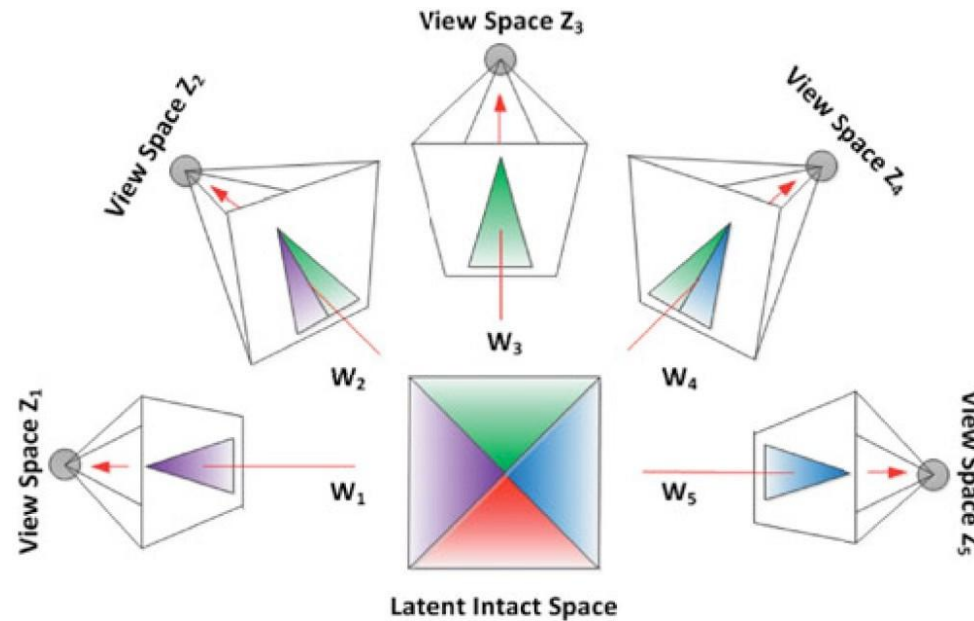
Figure 1: Schematic diagram of DNN-based multi-view representation learning approaches.

Deep Canonically Correlated Autoencoders (DCCAE)



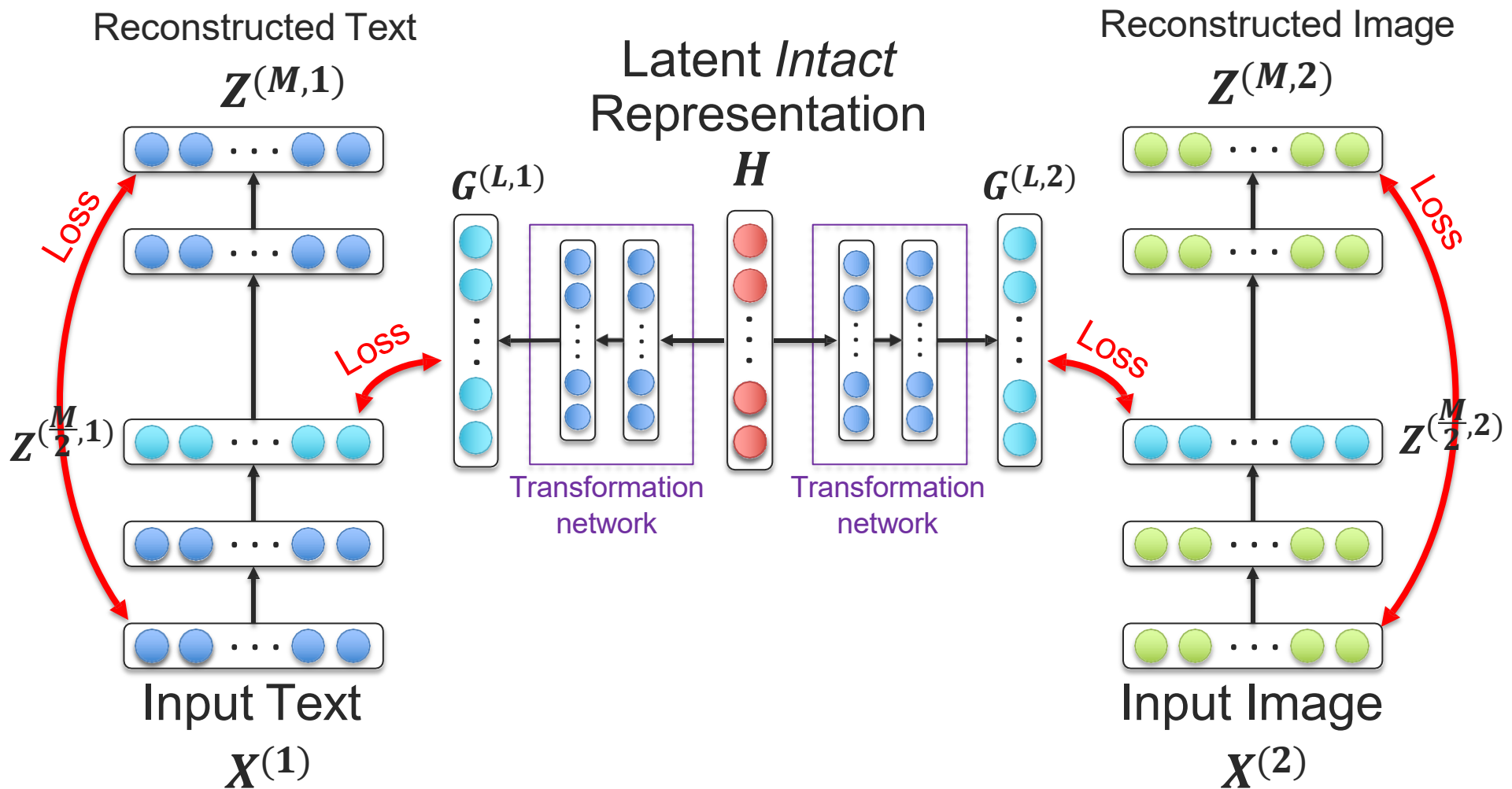
Multi-view Latent “Intact” Space

Given multiple views z_i from the same “object”:



- 1) There is an “intact” representation which is *complete* and *not damaged*
- 2) The views z_i are partial (and possibly degenerated) representations of the intact representation

Auto-Encoder in Auto-Encoder Network



Auto-Encoder in Auto-Encoder Network

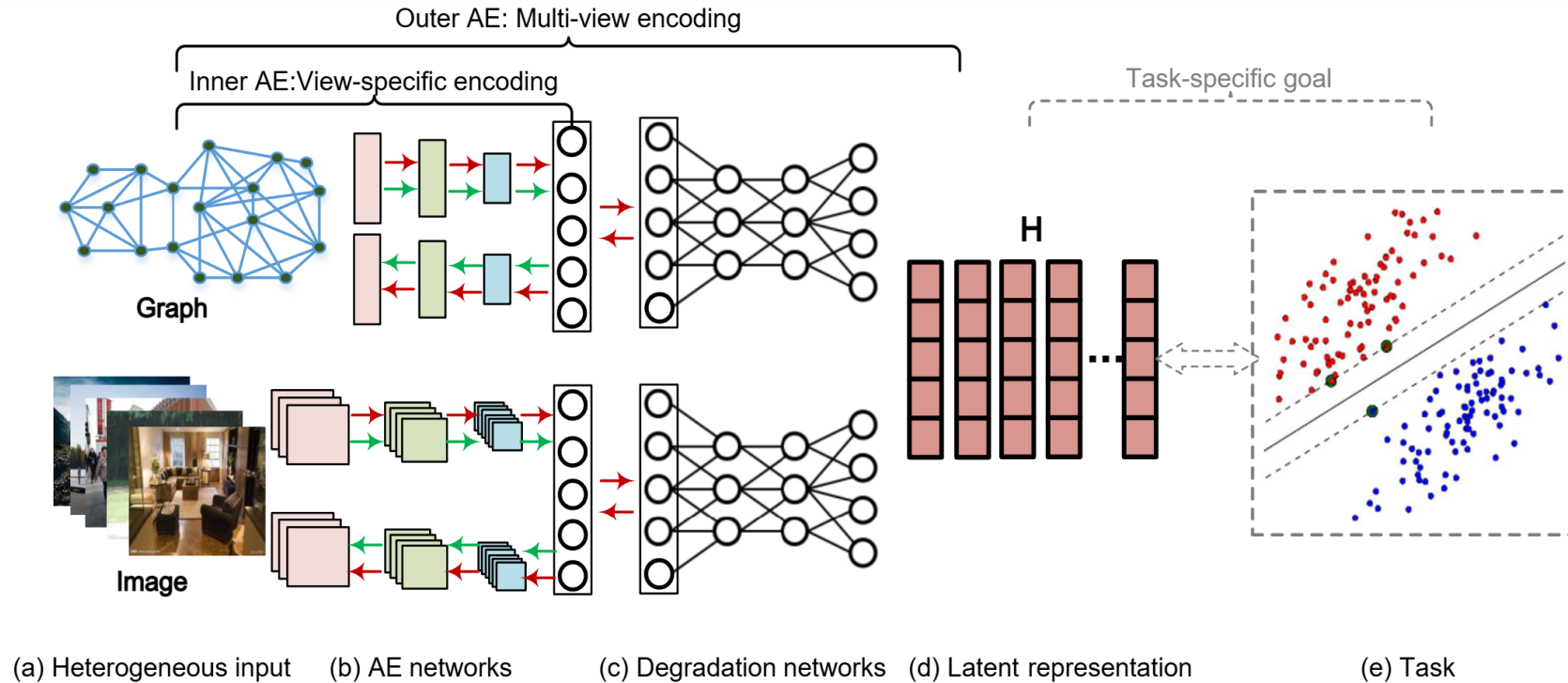
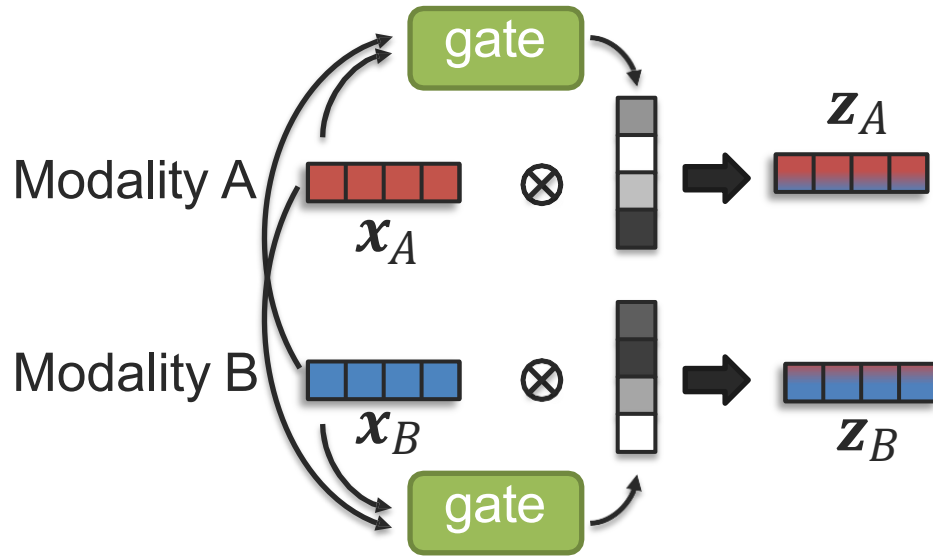


Figure 1: Overview of the Autoencoder in Autoencoder Networks (AE²-Nets). The key components are the nested autoencoder networks, which are composed of the inner AE networks (shown as the circle with **green arrows**) for *view-specific encoding* and the outer AE networks (shown as the circle with **red arrows**) for *multi-view encoding*. *View-specific encoding* automatically extracts features from each view while *multi-view encoding* ensures the intact latent representation can be mapped back to each view with *degradation process*. Accordingly, the intrinsic information from multiple views are encoded into the learned latent intact representation. The learned latent representation could be used for subsequent tasks, and the task-specific goal could flexibly be incorporated into our framework as well (shown in gray dash lines).

Gated Coordination



Gated coordination:

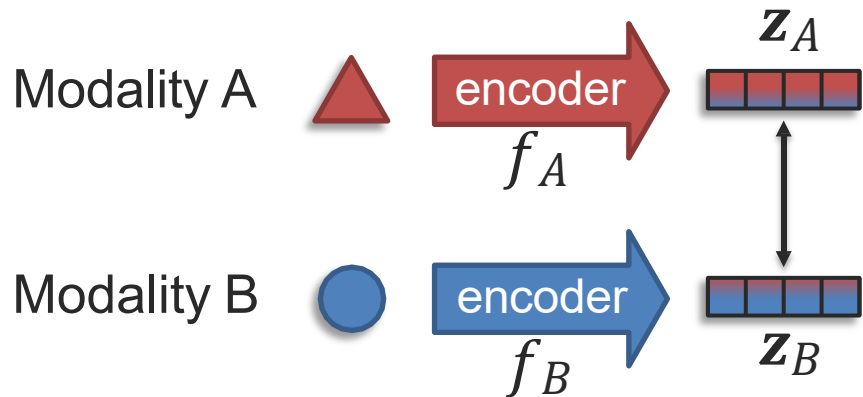
$$z_A = g_A(x_A, x_B) (x_A$$

$$z_B = g_B(x_A, x_B) (x_B$$

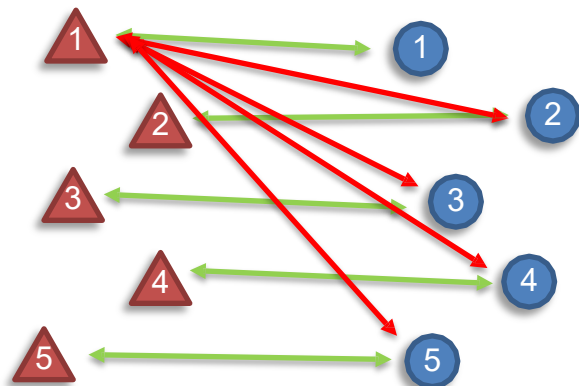
➡ Related to attention modules in transformers

More about it next week!

Coordination with Contrastive Learning



Paired data: $\{\triangle, \circ\}$
(e.g., images and text descriptions)



Contrastive loss:

→ brings **positive pairs** closer and pushes **negative pairs** apart

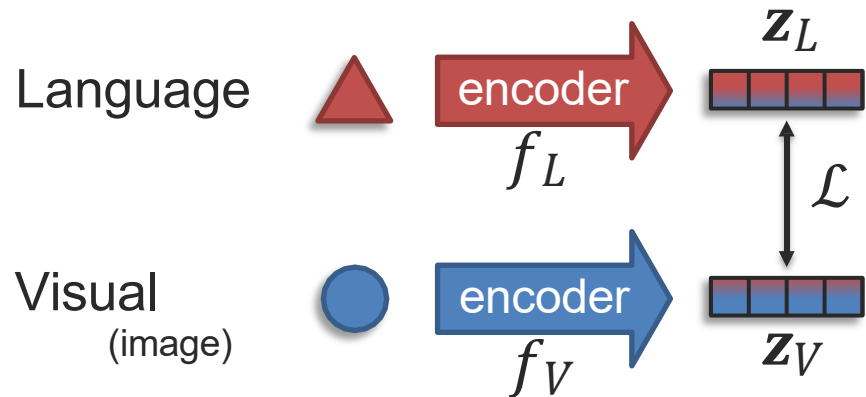
Simple contrastive loss:

$$\max\{0, \alpha + \underbrace{\text{sim}(z_A, z_B^+)}_{\text{positive pairs}} - \underbrace{\text{sim}(z_A, z_B^-)}_{\text{negative pair}}\}$$

Similarity functions are often cosine similarity

→ Similar to hinge loss

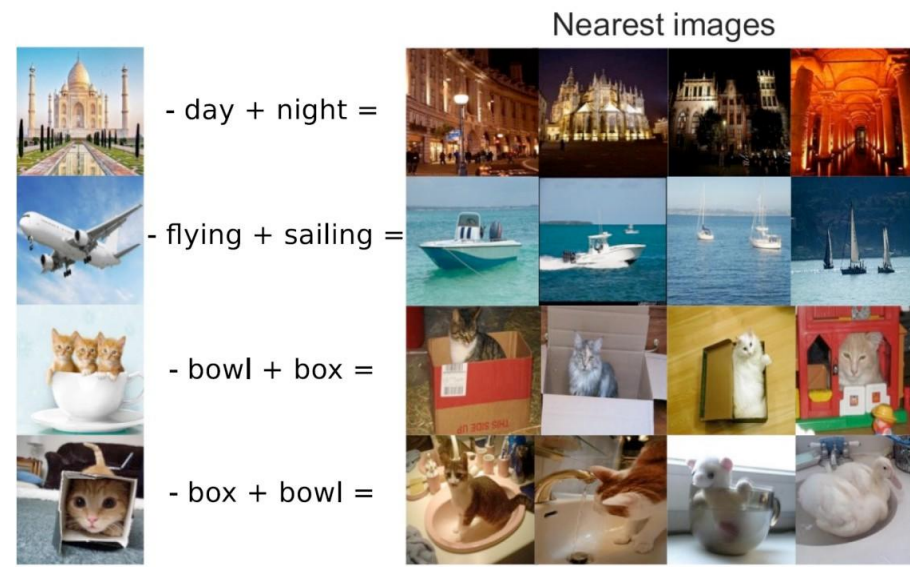
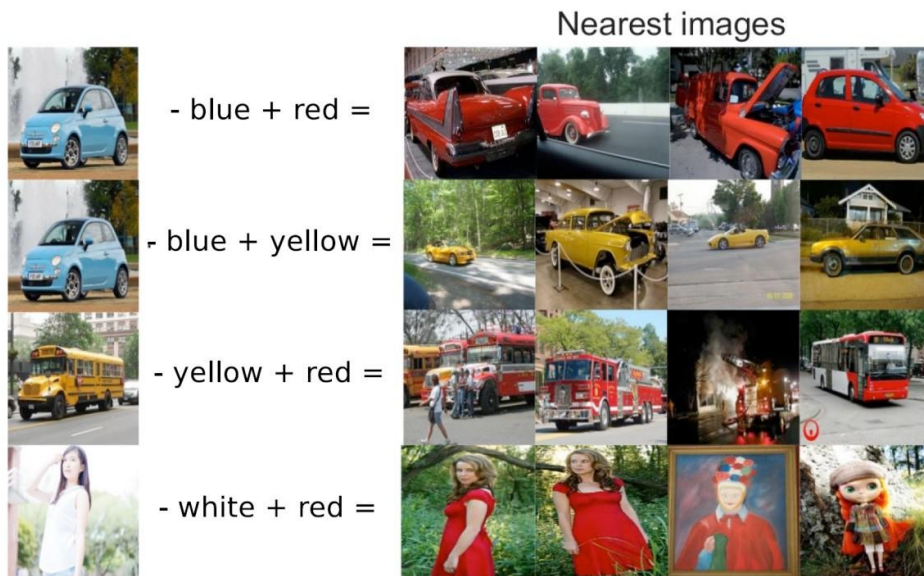
Example – Visual-Semantic Embeddings



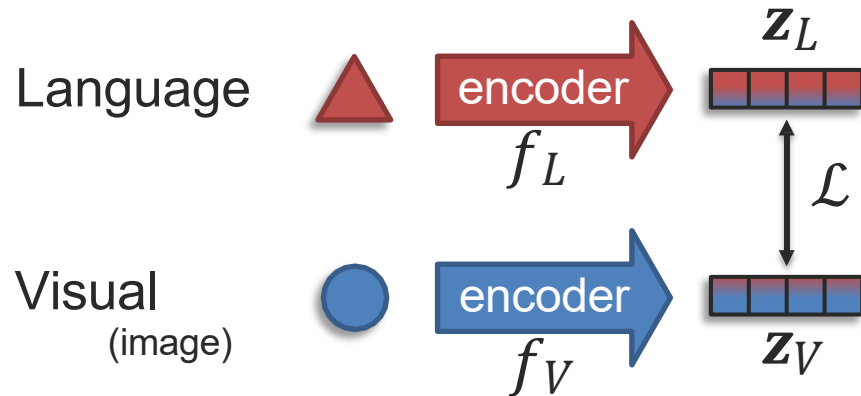
Two contrastive loss terms:

$$\max\{0, \alpha + \text{sim}(\mathbf{z}_L, \mathbf{z}_V^+) - \text{sim}(\mathbf{z}_L, \mathbf{z}_V^-)\}$$

$$+ \max\{0, \alpha + \text{sim}(\mathbf{z}_V, \mathbf{z}_L^+) - \text{sim}(\mathbf{z}_V, \mathbf{z}_L^-)\}$$



Example – CLIP (Contrastive Language–Image Pre-training)



Popular contrastive loss: InfoNCE

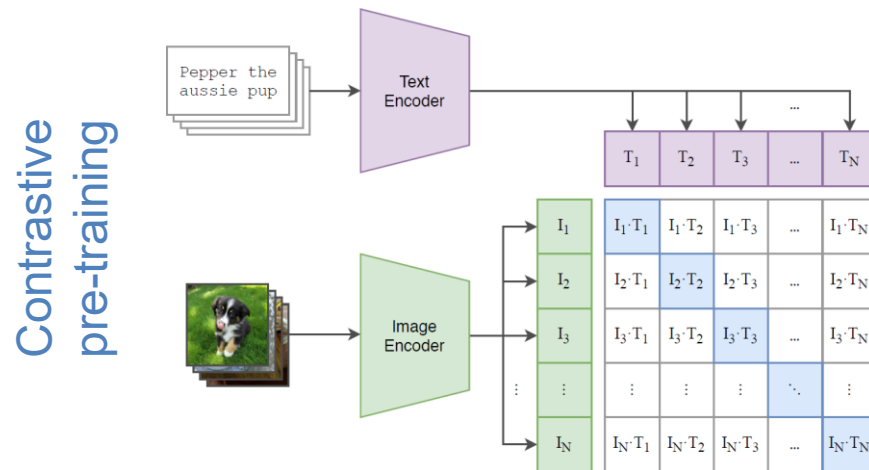
$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\text{sim}(z_A^i, z_B^i)}{\sum_{j=1}^N \text{sim}(z_A^i, z_B^j)}$$

positive pairs

negative pairs and positive pairs

Similarity function can be cosine similarity

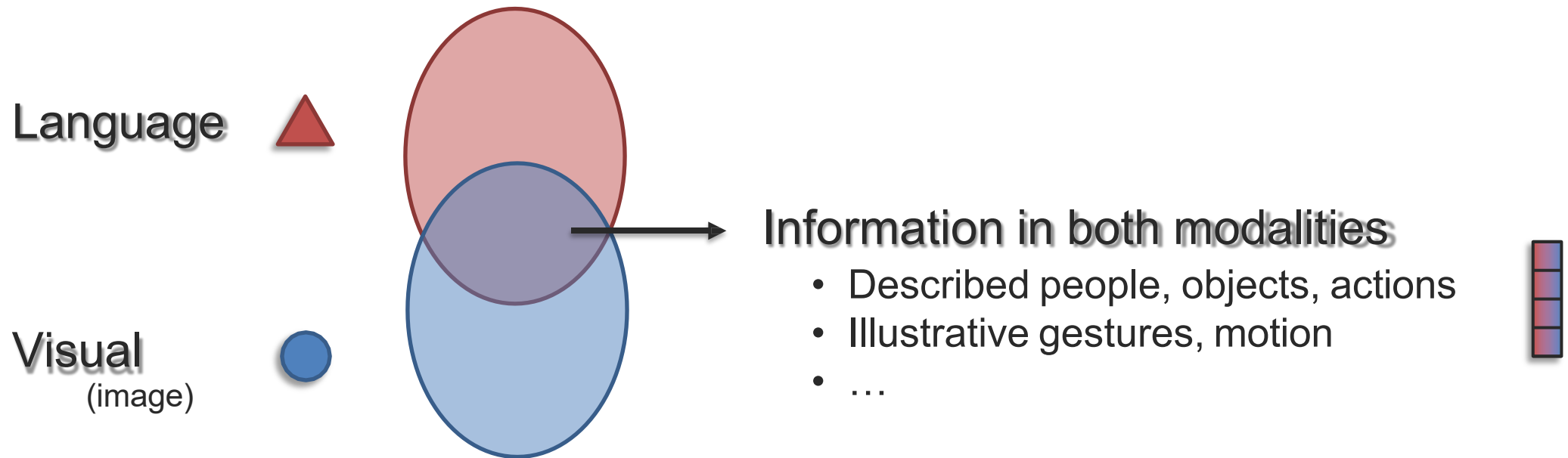
Positive and negative pairs:



CLIP encoders (f_L and f_V) are great for language-vision tasks

z_L and z_V are coordinated but not identical representation spaces

Multimodal Coordination – Information Theory

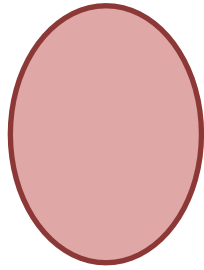


Information and Entropy – Information Theory

Language



x



How much information in the modality?

Information Theory (Shannon, 1948)

Main intuition: “Information value” of a communicated message x depends on how random its content is

x : “1,1,1,1,1,1,1,1,1,1,1,1,1”

➡ Not very random... So, low information

x : “0,1,0,1,0,0,1,1,1,0,0,1”

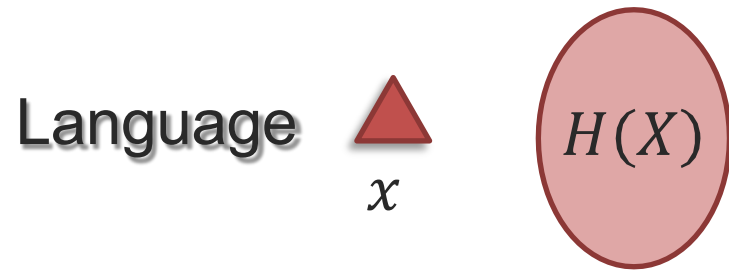
➡ More random... So, higher information

Information content $I(x)$

$$I(x) \sim \frac{1}{p(x)}$$

$$I(x) = \log \left(\frac{1}{p(x)} \right) = -\log(p(x))$$

Information and Entropy – Information Theory



How much information in the modality?

Information Theory (Shannon, 1948)

Information content $I(X) = -\log(p(X))$

➡ For discrete alphabet $\mathcal{X}$, then X is discrete random variable

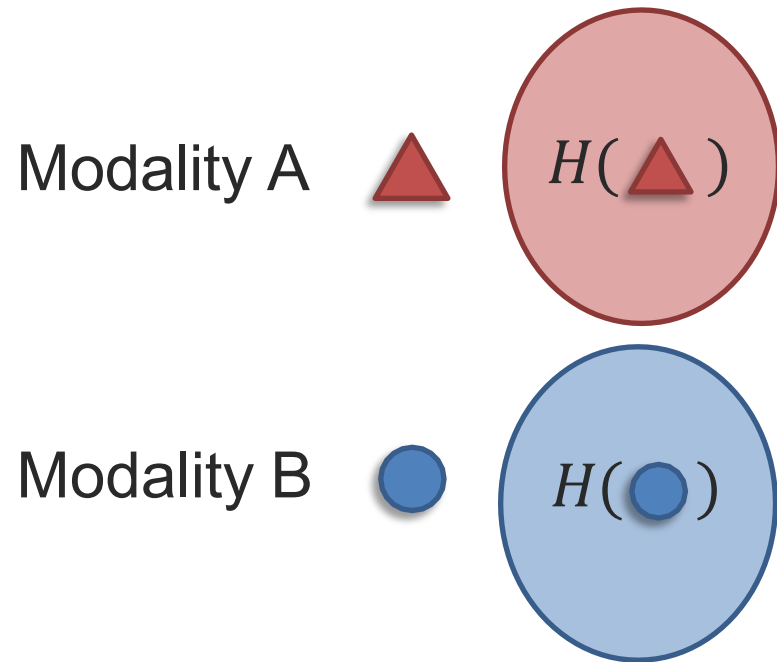
Entropy: weighted average of all possible outcomes from $\mathcal{X}$

$$H(X) = E[I(X)] = E[-\log(p(X))] = - \sum_{x \in \mathcal{X}} p(X) \log(p(X))$$

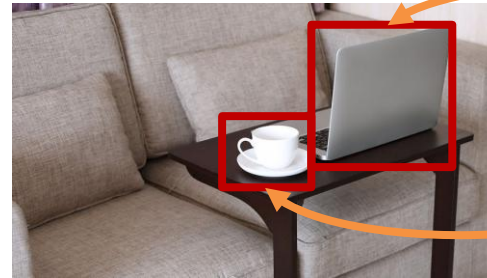
➡ Entropy can also be defined for continuous random variables

Entropy with Two Modalities

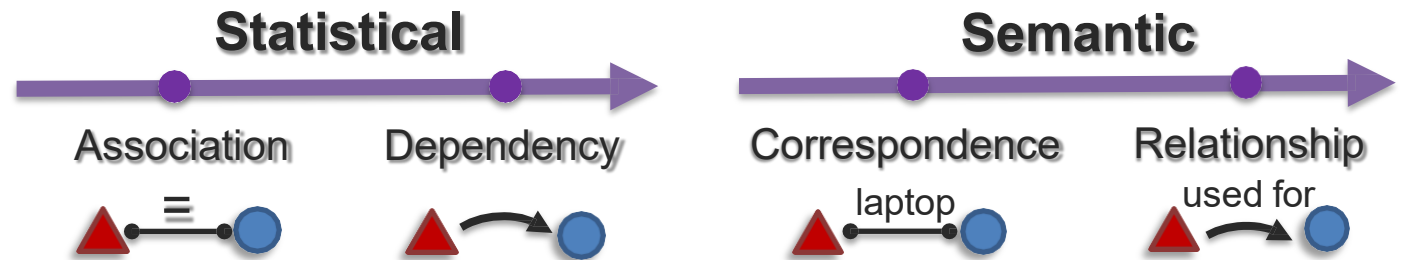
If no overlapping
information



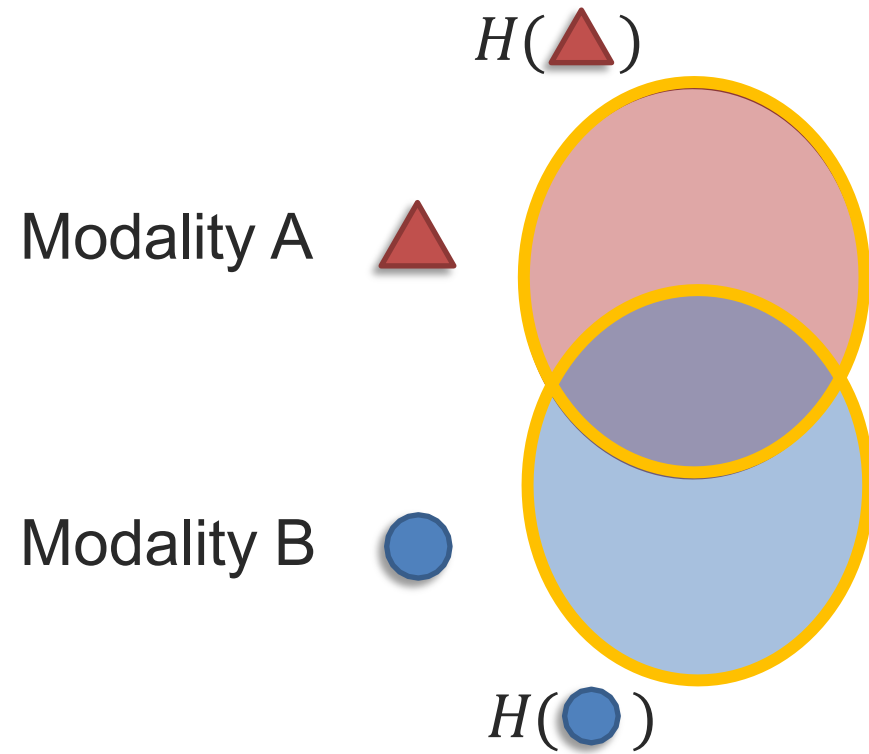
➡ But in most real-world scenarios,
modalities are *inter-connected*



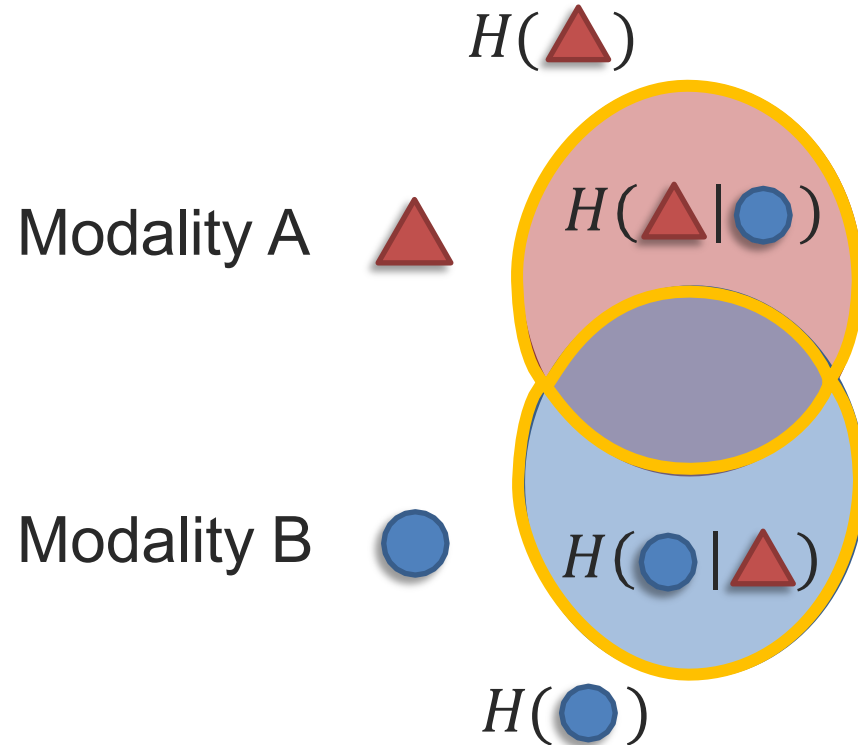
A **teacup** on the right of a **laptop**
in a clean room.



Entropy with Two Modalities



Entropy with Two Modalities



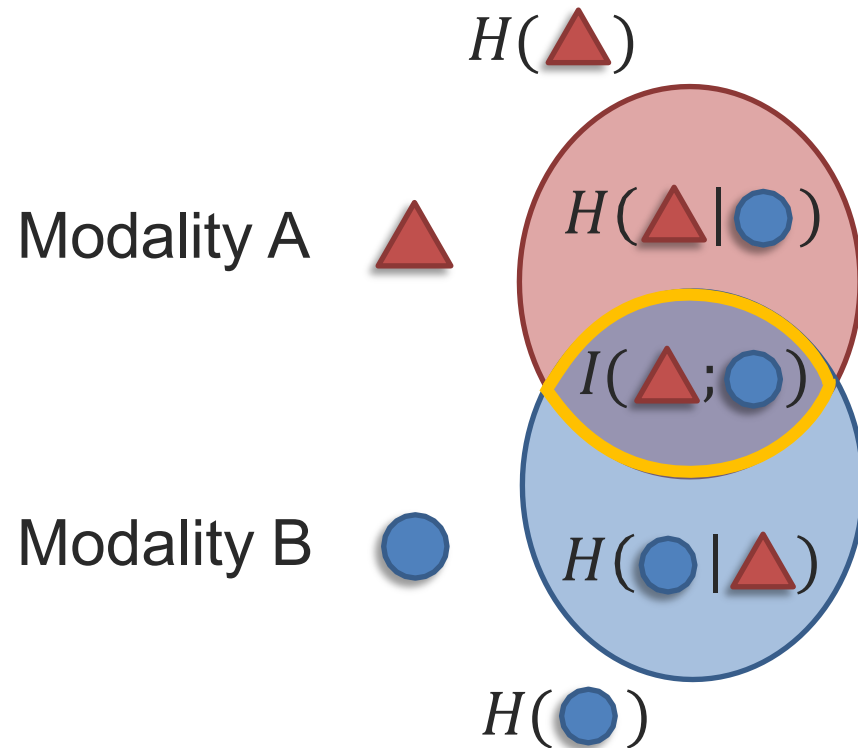
Conditional entropy $H(Y|X)$

$$H(Y|X) = -E_{X,F} [\log p(y|x)]$$

$$= -E_{X,F} \left[\log \frac{p(x, y)}{p(x)} \right]$$

If X and Y independent, $H(Y|X) = H(Y)$.
If X fully determines Y , then $H(Y|X) = 0$.

Entropy with Two Modalities



Mutual information $I(X; Y)$

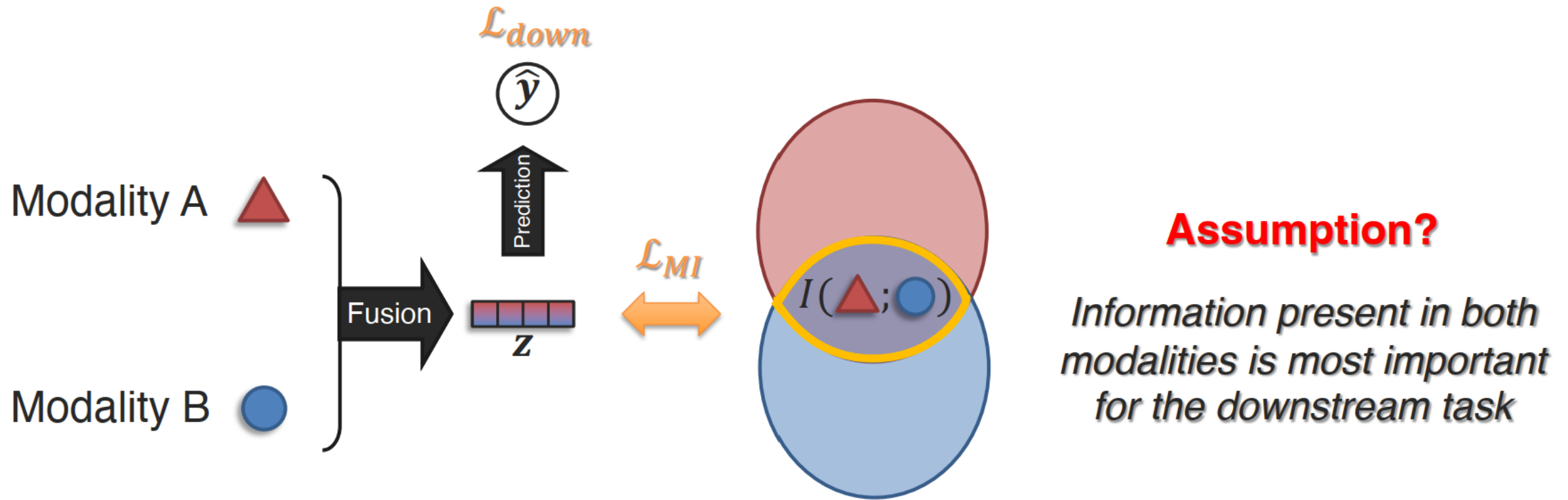
$$I(X; Y) = H(X) - H(X|Y)$$

$$= E_{X,Y} \left[\log \frac{1}{P_X(x)} + \log \frac{P_{XY}(x, y)}{P_Y(y)} \right]$$

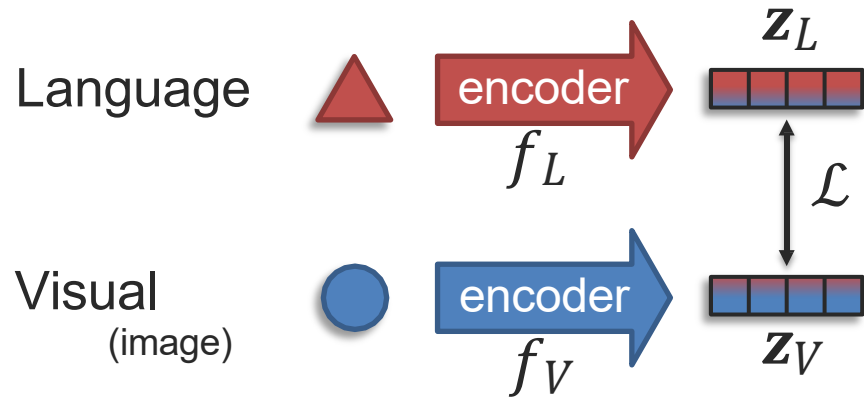
$$I(X; Y) = E_{X,Y} \left[\log \frac{P_{XY}(x, y)}{P_X(x)P_Y(y)} \right]$$

using KL-divergence $\leftarrow I(X; Y) = D_{KL}(P_{XY}(x, y) \parallel P_X(x)P_Y(y))$

Multimodal Fusion with Mutual Information



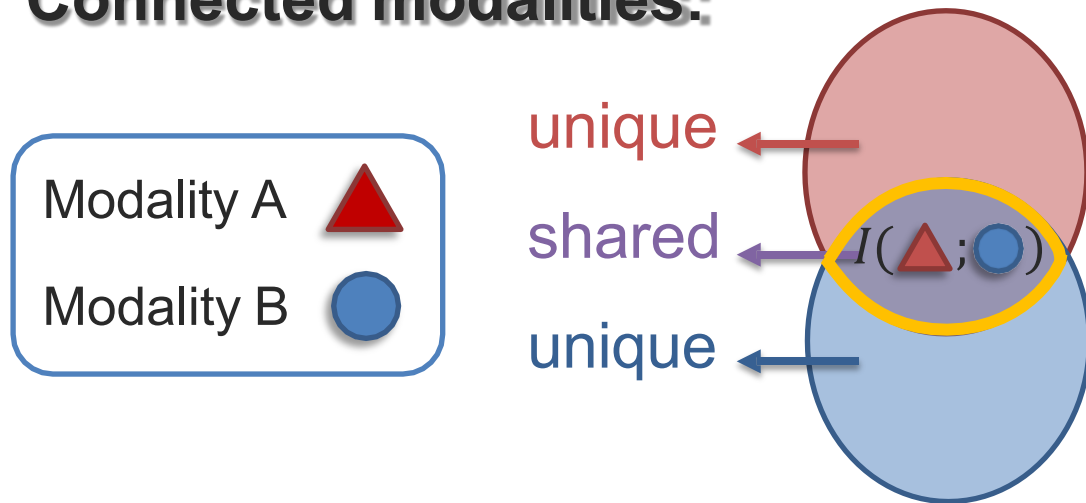
Contrastive Learning and Connected Modalities



Popular contrastive loss: InfoNCE

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\text{sim}(z_A^i, z_B^i)}{\sum_{j=1}^N \text{sim}(z_A^i, z_B^j)}$$

Connected modalities:



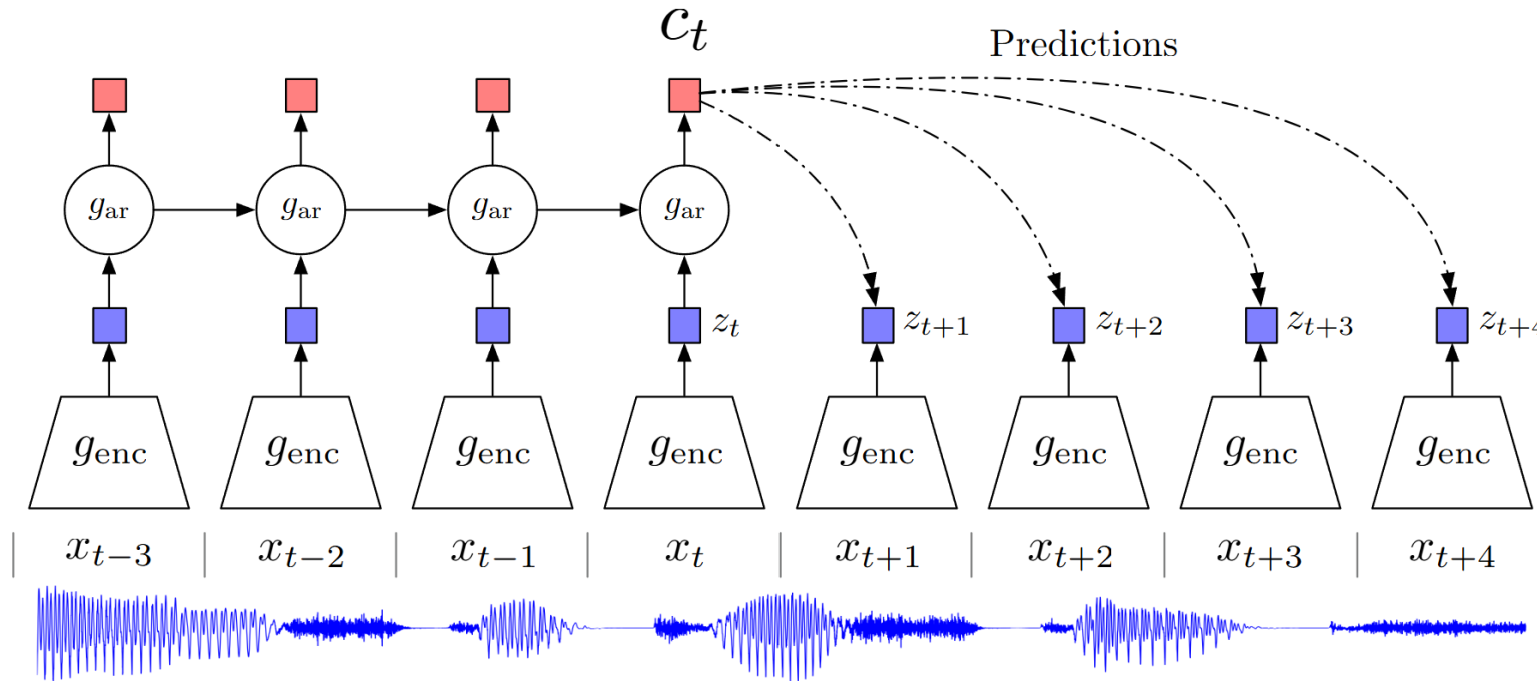
Mutual information $I(X; Y)$

$$E_{X,Y} \left[\log \frac{P_{XY}(x, y)}{P_X(x)P_Y(y)} \right]$$



CLIP focuses on shared connections

Contrastive Learning and Connected Modalities

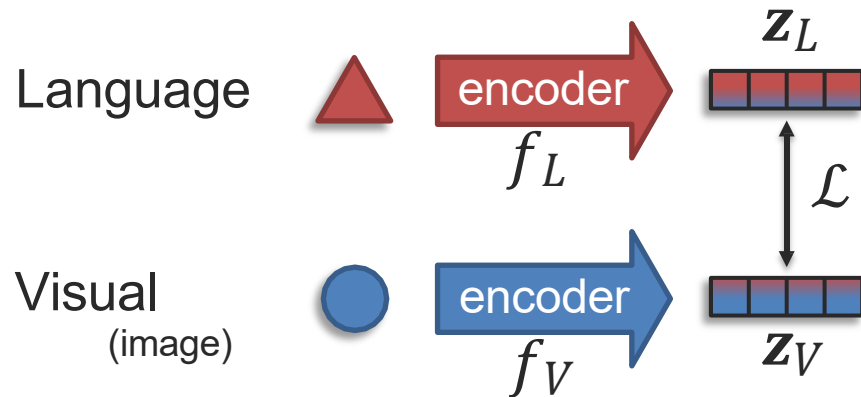


1. **Encoder (g_{enc}):** This maps the input sequence of observations x_t into a sequence of latent representations z_t . The encoder reduces the dimensionality of the input data, making it easier to model conditional predictions.
2. **Autoregressive Model (g_{ar}):** This summarizes all $z_{\leq t}$ in the latent space to produce a context latent representation c_t . The autoregressive model captures temporal dependencies and provides a context for predicting future representations.
3. **Contrastive Loss (InfoNCE):** The model is trained using a probabilistic contrastive loss based on NCE. This loss function encourages the model to distinguish between positive samples (future observations given the context) and negative samples (randomly sampled observations). The InfoNCE loss is defined as:

$$\mathcal{L}_N = -\mathbb{E}_X \left[\log \frac{f_k(x_{t+k}, c_t)}{\sum_{x_j \in X} f_k(x_j, c_t)} \right]$$

where $f_k(x_{t+k}, c_t)$ is a density ratio proportional to $\frac{p(x_{t+k}|c_t)}{p(x_{t+k})}$.

Contrastive Learning and Mutual Information



InfoNCE/CL:

- 'Captures' mutual information
- Optimizes a lower bound on mutual information

In other words:

InfoNCE:

$$\mathcal{L} = -\mathbb{E} \left[\log \frac{f(\mathbf{x}_A^i, \mathbf{x}_B^i)}{\sum_{j=1}^N f(\mathbf{x}_A^i, \mathbf{x}_B^j)} \right] \quad \text{critic function}$$

Critic function f is trained to be a binary classifier distinguishing $\mathbf{x}_A, \mathbf{x}_B \sim p(\mathbf{x}_A, \mathbf{x}_B)$ vs $\mathbf{x}_A, \mathbf{x}_B \sim p(\mathbf{x}_A)p(\mathbf{x}_B)$

At optimal loss, $f^*(\mathbf{x}_A, \mathbf{x}_B) = \frac{p(\mathbf{x}_\#, \mathbf{x}_B)}{p(\mathbf{x}_\#)p(\mathbf{x}_B)}$.

Plugging f^* back into $\mathcal{L}$ gives:

$$\mathcal{L}^* \geq \mathbb{E} \left[\log \frac{p(\mathbf{x}_A)p(\mathbf{x}_B)}{p(\mathbf{x}_A, \mathbf{x}_B)} N \right] = -I(X_A, X_B) + \log N$$

$$I(X_A, X_B) \geq \log N - \mathcal{L}^*$$

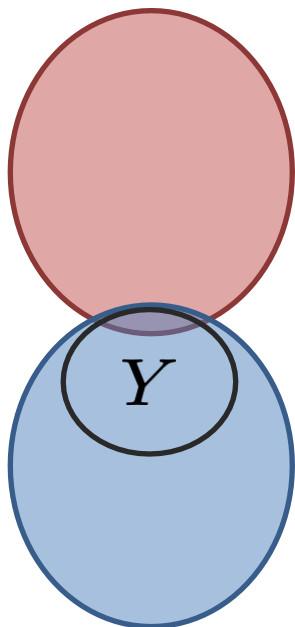
Multiview Redundancy and Contrastive Learning

Open
challenges

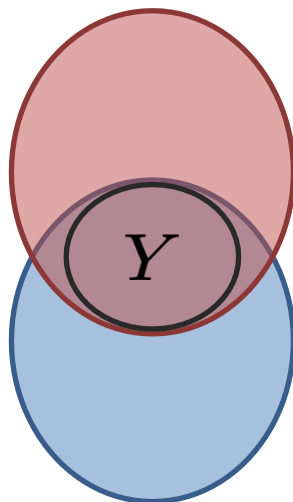
How much information should be shared?

Multi-view redundancy: $I(X_1; X_2) = I(X_1; Y)$

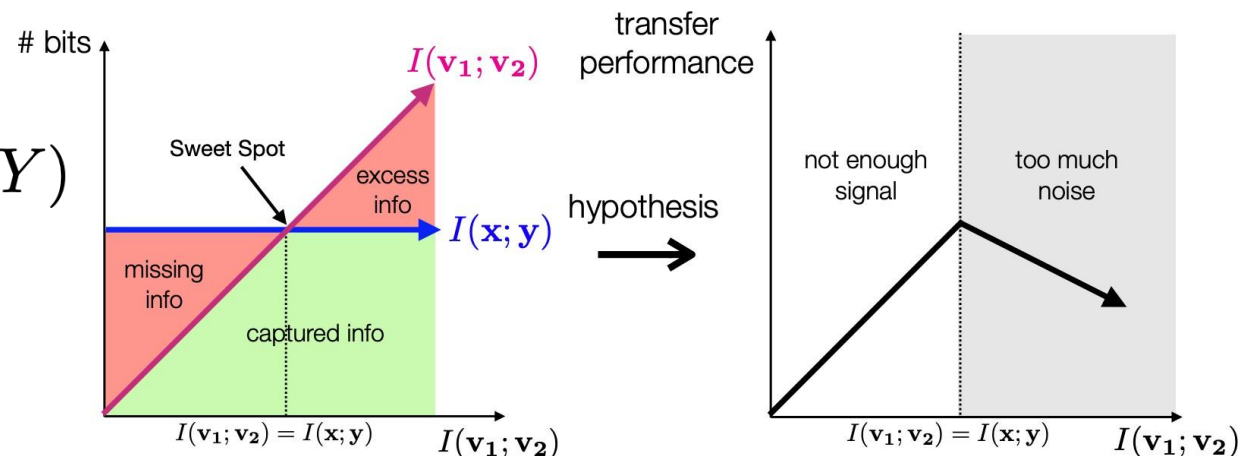
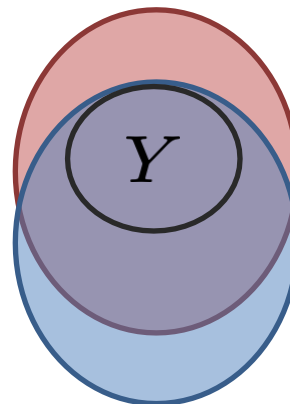
Not enough signal



Just right



Too much noise



Multi-view redundancy
may not hold for
multimodal problems!

[Tian et al., What makes for Good Views for Contrastive Learning? NeurIPS 2020]

[Tosh et al., Contrastive Learning, Multi-view Redundancy, and Linear models. ALT 2021]

Multiview Redundancy and Contrastive Learning

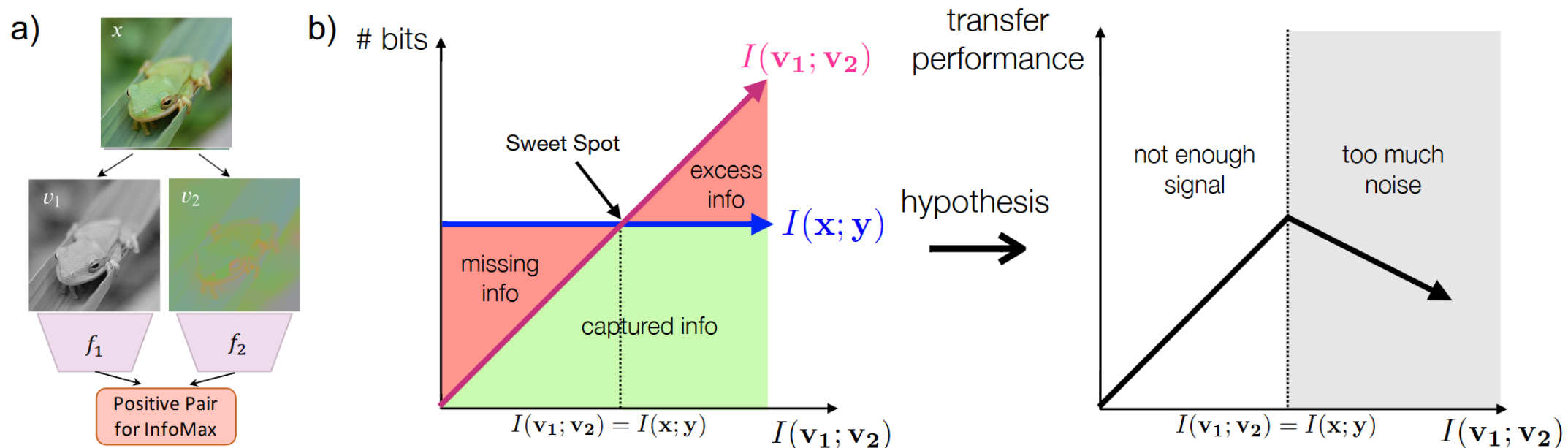
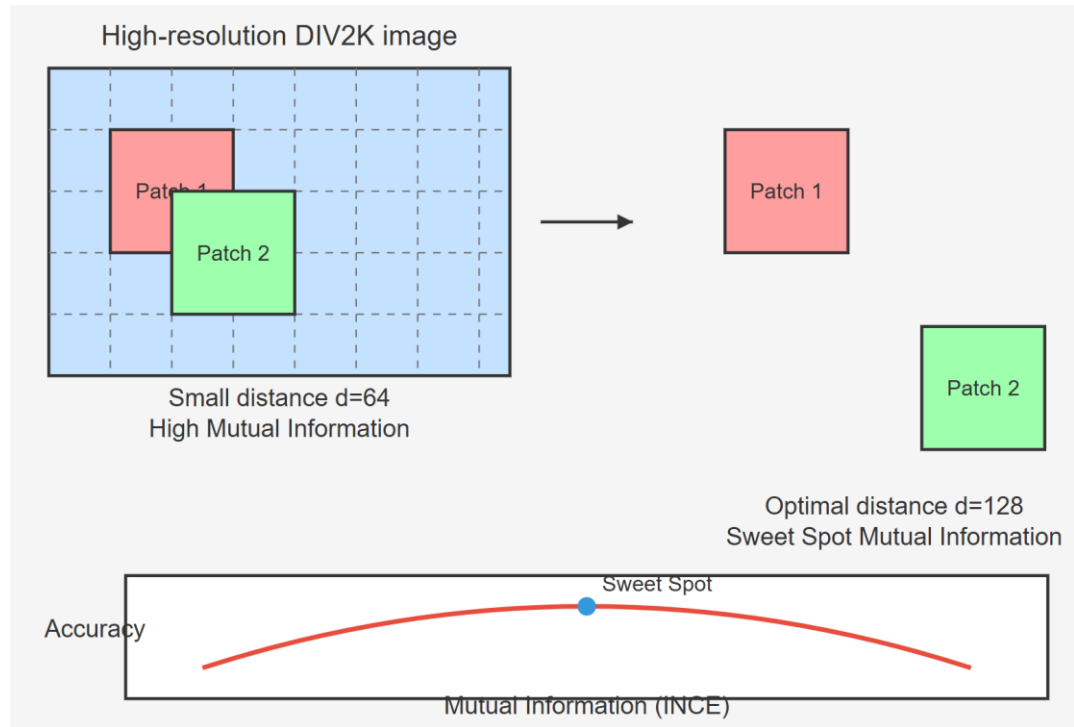


Figure 1: **(a)** Schematic of multiview contrastive representation learning, where an image is split into two views, and passed through two encoders to learn an embedding where the views are close relative to views from other images. **(b)** When we have views that maximize $I(\mathbf{v}_1; \mathbf{y})$ and $I(\mathbf{v}_2; \mathbf{y})$ (how much task-relevant information is contained) while minimizing $I(\mathbf{v}_1; \mathbf{v}_2)$ (information shared between views, including both task-relevant and irrelevant information), there are three regimes: *missing information* which leads to degraded performance due to $I(\mathbf{v}_1; \mathbf{v}_2) < I(\mathbf{x}; \mathbf{y})$; *excess noise* which worsens generalization due to additional noise; *sweet spot* where the only information shared between $\mathbf{v}_1$ and $\mathbf{v}_2$ is task-relevant and such information is complete.

Multiview Redundancy and Contrastive Learning

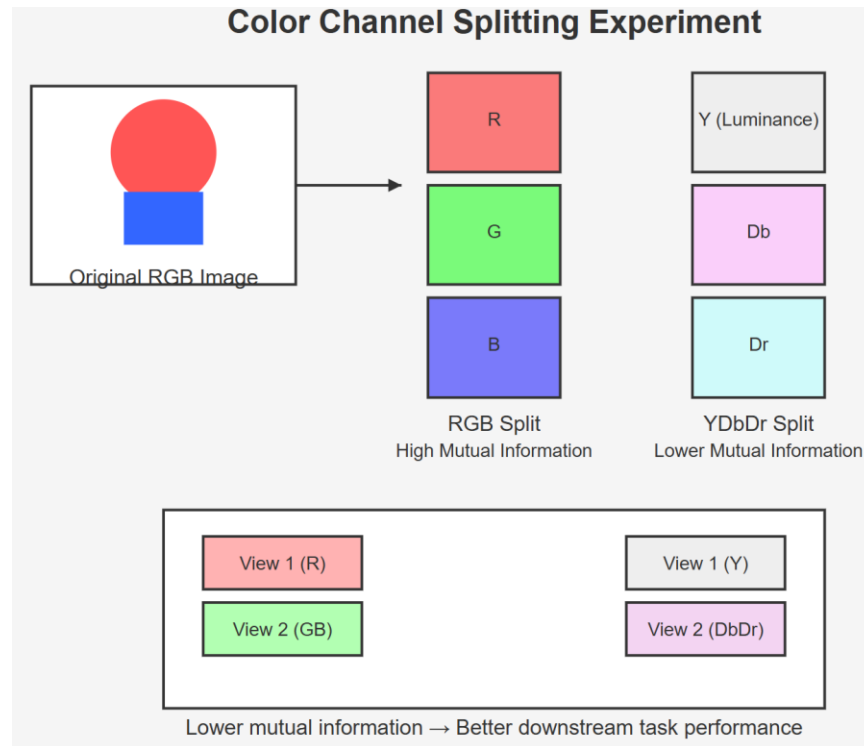
Spatial Patches Experiment



This experiment demonstrates that as the distance between image patches increases, the mutual information decreases, affecting downstream performance. The researchers found a "sweet spot" at distance $d=128$ where accuracy peaks, forming a reverse U-shaped curve on both STL-10 and CIFAR-10 classification tasks.

Multiview Redundancy and Contrastive Learning

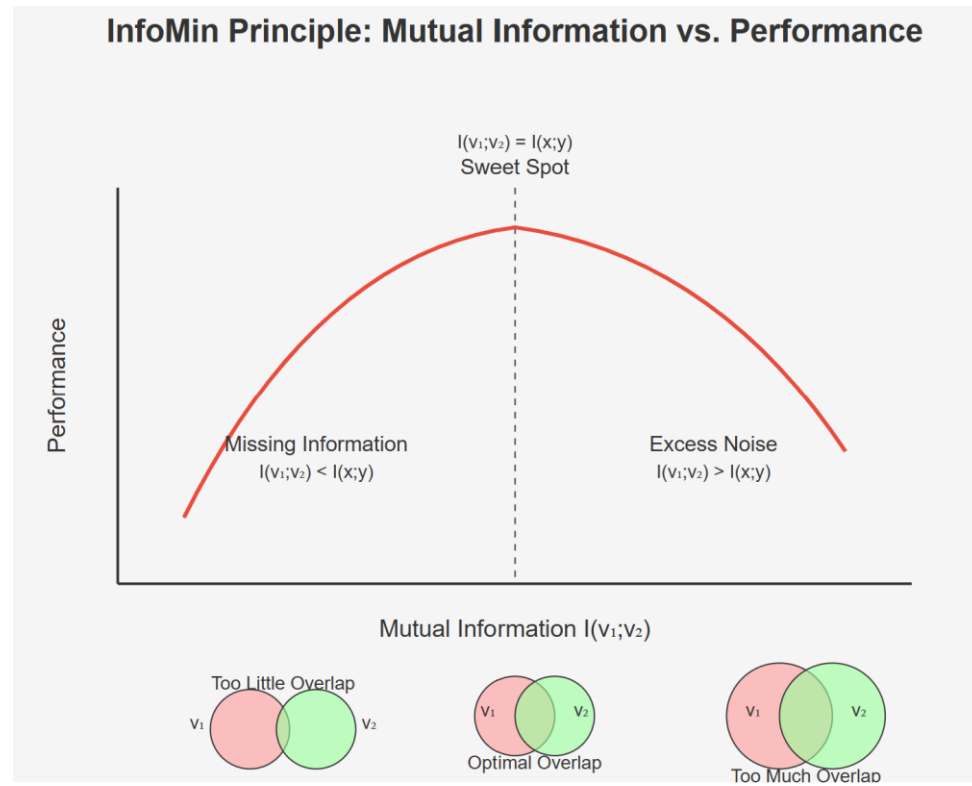
Color Channel Splitting Experiment



By splitting images into different color channels (RGB vs. YDbDr), the researchers showed that the luminance-chrominance decomposition (YDbDr) provides views with lower mutual information, which consistently improves downstream performance on both classification and segmentation tasks.

Multiview Redundancy and Contrastive Learning

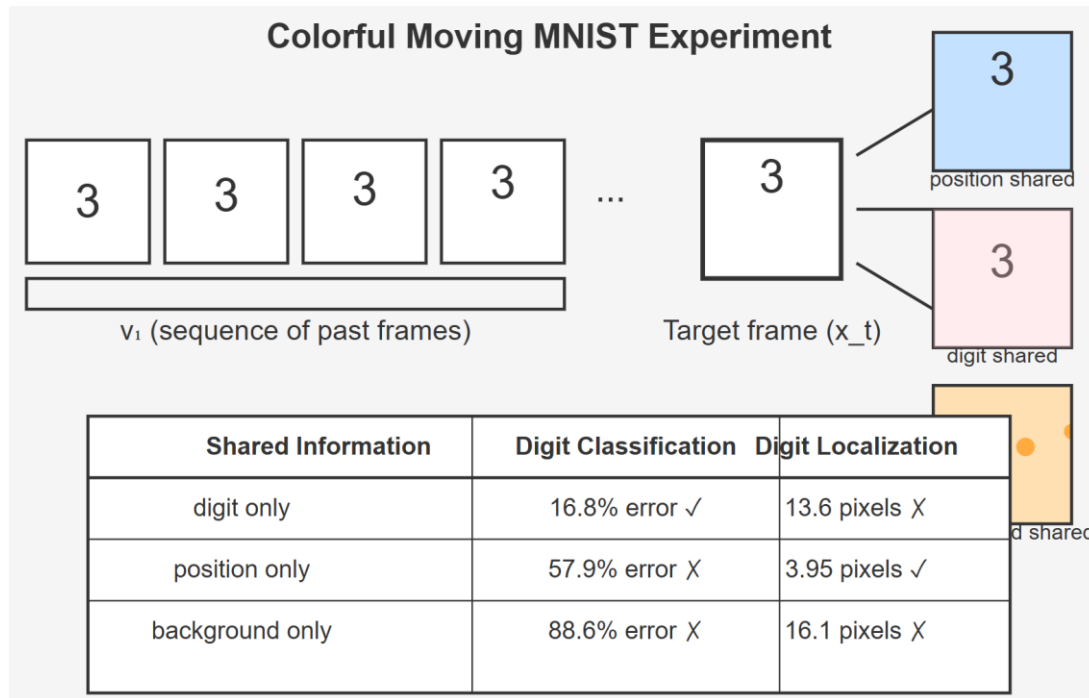
InfoMin Principle and MI-Performance Curve



This visualization illustrates the core concept of the InfoMin principle: the optimal views for contrastive learning share just enough information about the downstream task while discarding nuisance variables. The performance curve forms a reverse U-shape with a sweet spot where $I(v_1;v_2) = I(x;y)$.

Multiview Redundancy and Contrastive Learning

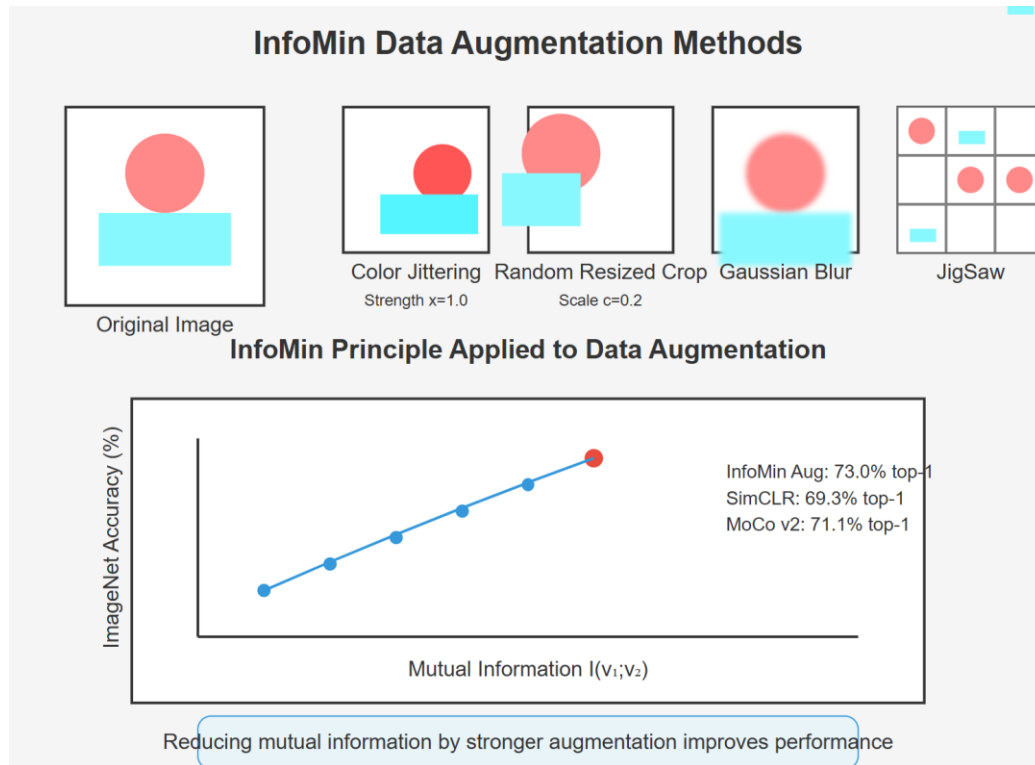
Colorful Moving MNIST Experiment



This toy experiment combines Moving-MNIST with STL-10 backgrounds to demonstrate that optimal views depend on the downstream task. When only digit information is shared between views, digit classification improves but localization suffers. When only position information is shared, localization improves but classification suffers.

Multiview Redundancy and Contrastive Learning

InfoMin Data Augmentation Methods

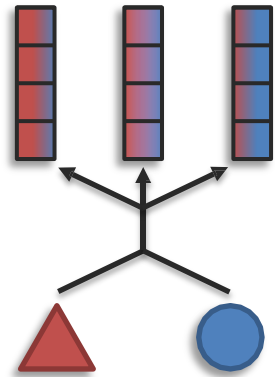


The researchers systematically analyzed various data augmentation techniques (Color Jittering, Random Resized Crop, Gaussian Blur, RandAugment, and JigSaw) and showed that increasingly stronger augmentations further reduce mutual information between views, leading to improved ImageNet linear readout performance up to 73.0% top-1 accuracy.

Representation

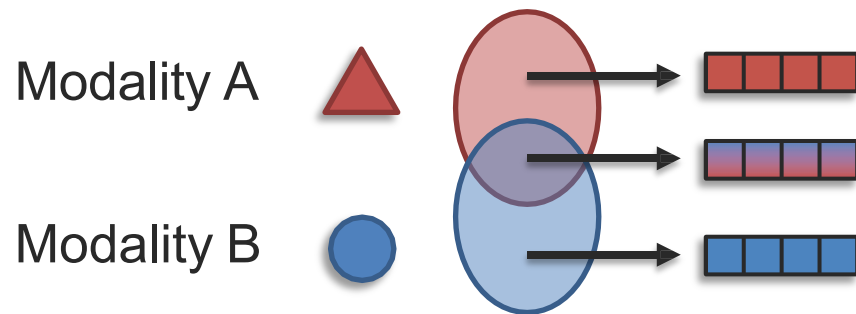
Fission

Sub-Challenge 1c: Representation Fission

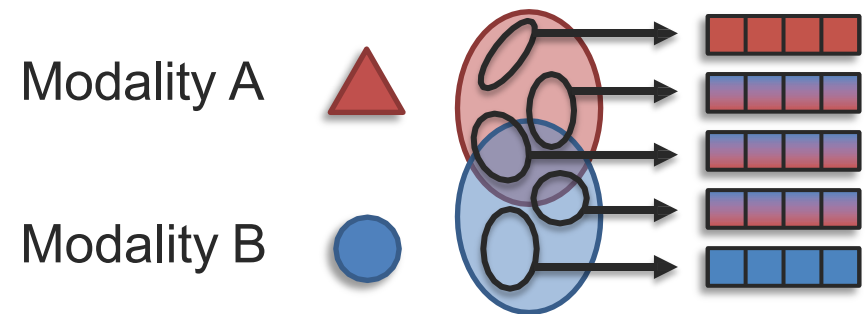


Definition: Learning a new set of representations that reflects multimodal internal structure such as data factorization or clustering

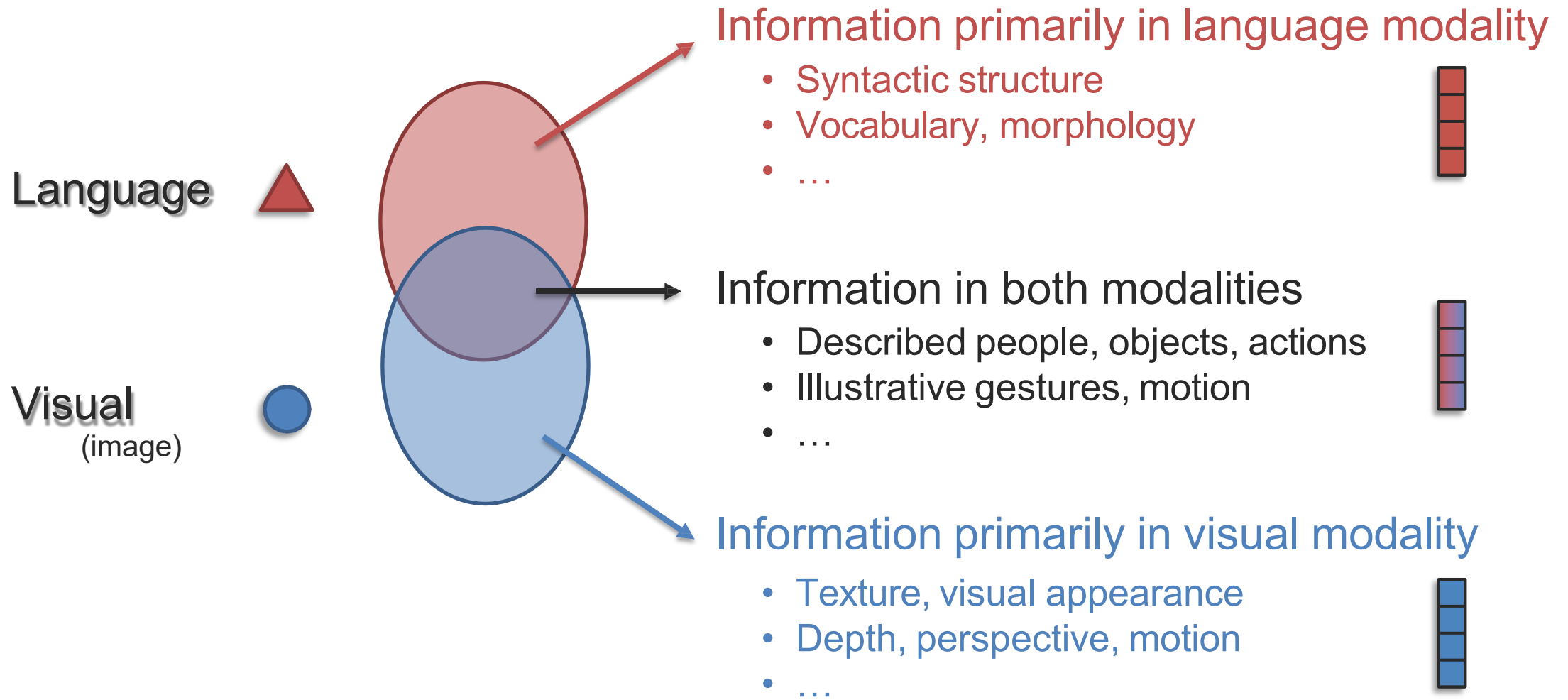
Modality-level fission:



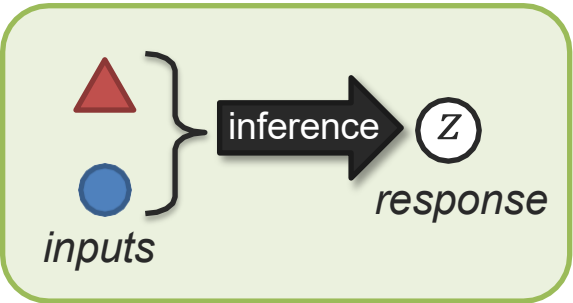
Fine-grained fission:



Modality-Level Fission



Recall Taxonomy of Interactions



Multimodal Communication



Redundancy

signal	response
a	→ □
b	→ □

signal	response
a+b	→ □
a+b	→ □

Equivalence
Enhancement

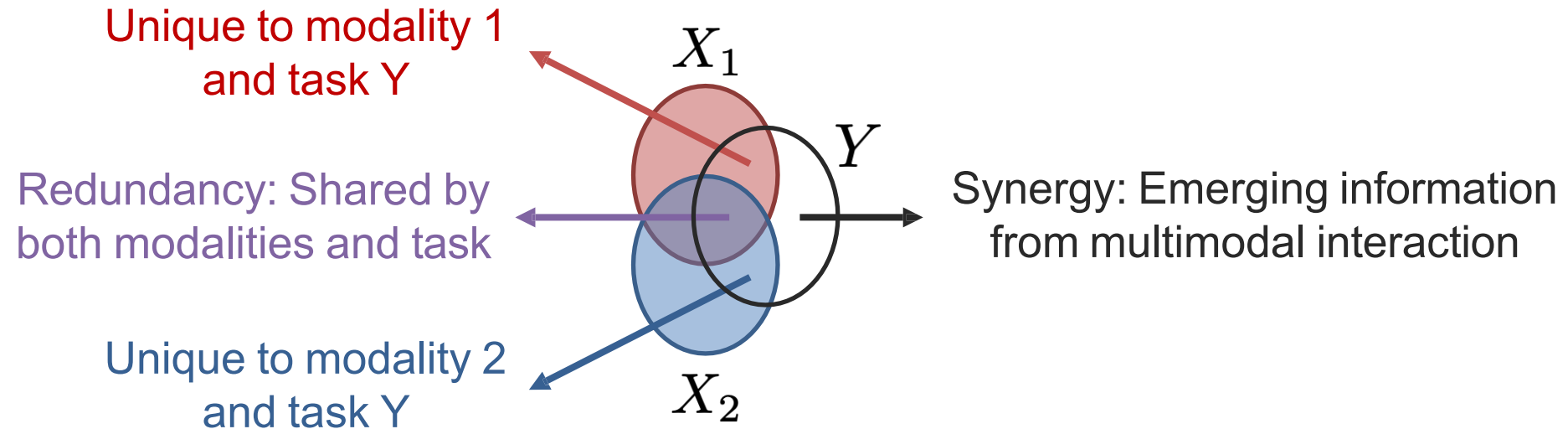
Nonredundancy

a	→ □
b	→ ○

a+b	→ □ and ○
a+b	→ □
a+b	→ □ (or □)
a+b	→ △

Independence
Dominance
Modulation
Emergence

Representation Fission via Information Theory

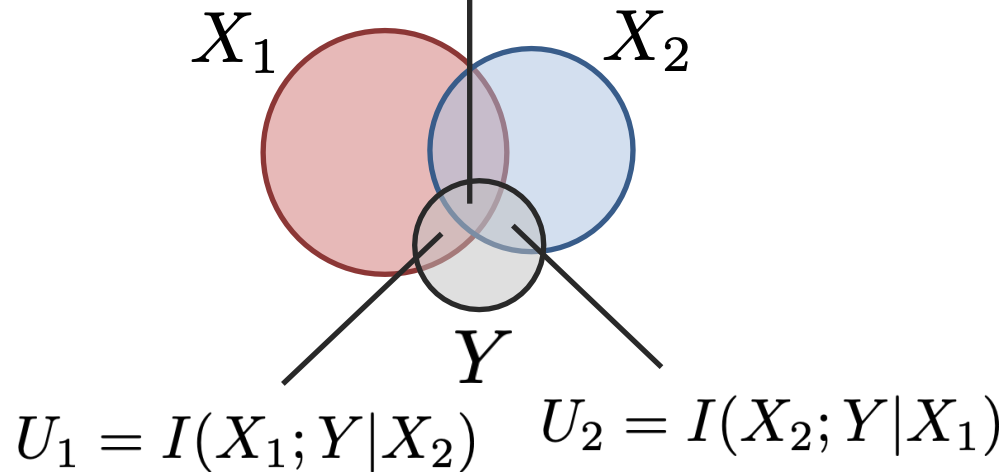


Partial Information Decomposition

Classical Information Theory

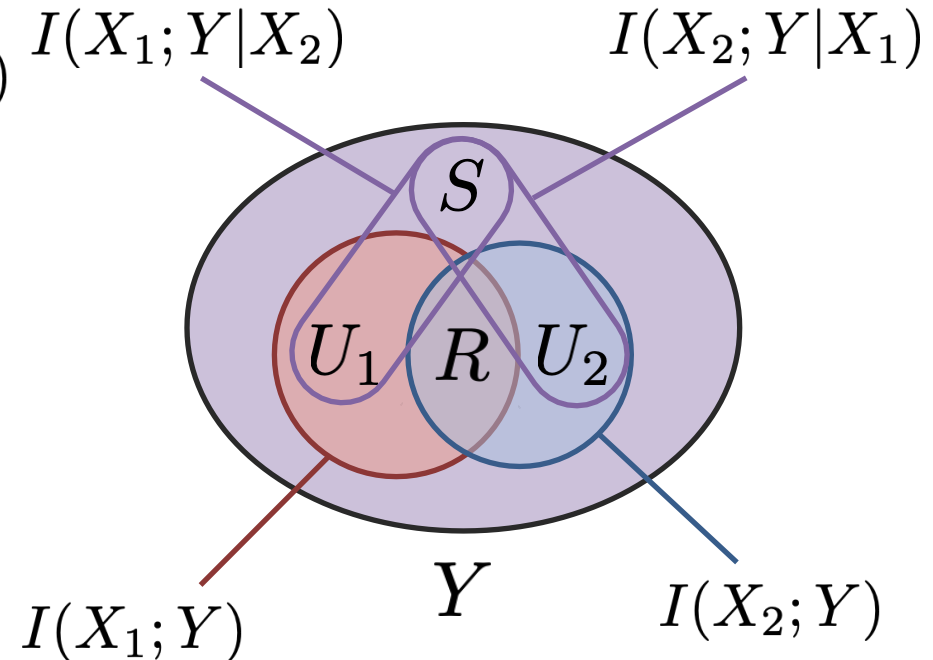
$$R = I(X_1; X_2; Y) = I(X_1; X_2) - I(X_1; X_2|Y)$$

Can be negative!



No synergy!

Partial Information Decomposition



$$R - S = I(X_1; X_2; Y)$$

Explains negative!

$$R + U_1 + U_2 + S = I(X_1, X_2; Y)$$

Task-relevant multimodal info

Partial Information Decomposition

The paper introduces a new perspective on multivariate information theory by proposing a novel definition of redundancy. This definition aims to quantify the minimum information that any source provides about each possible outcome of a variable, averaged over all outcomes. Here's a detailed explanation:

New Definition of Redundancy

The authors define redundancy as the expected value of the minimum information that any source provides about each outcome of a variable S . This is formalized as:

$$I_{\min}(S; \{\mathbf{A}_1, \mathbf{A}_2, \dots, \mathbf{A}_k\}) = \sum_s p(s) \min_{\mathbf{A}_i} I(S = s; \mathbf{A}_i)$$

Where:

- $I(S = s; \mathbf{A}_i)$ is the specific information that source $\mathbf{A}_i$ provides about the outcome s of S .
- $p(s)$ is the probability of outcome s .
- $\min_{\mathbf{A}_i} I(S = s; \mathbf{A}_i)$ is the minimum information provided by any source $\mathbf{A}_i$ about outcome s .

Key Properties of $I_{\min}$

1. **Nonnegativity:** $I_{\min}$ is always nonnegative, which aligns with the intuitive understanding of information as a nonnegative quantity ≥ 0 .
2. **Upper Bound:** The redundant information is bounded by the information provided by each source, meaning $I_{\min} \leq I(S; \mathbf{A}_i)$ for all $\mathbf{A}_i \geq 0$.
3. **Maximal Redundancy:** The redundancy is maximized when all sources provide the same information about S , similar to how mutual information is maximized by self-information ≥ 0 .

This new definition of redundancy helps in decomposing multivariate information into meaningful components such as unique information, redundancy, and synergy. Unlike interaction information, which can be negative and difficult to interpret, this approach ensures that all components of information are nonnegative and provide clear insights into how different sources contribute to the overall information about a variable

Partial Information Decomposition

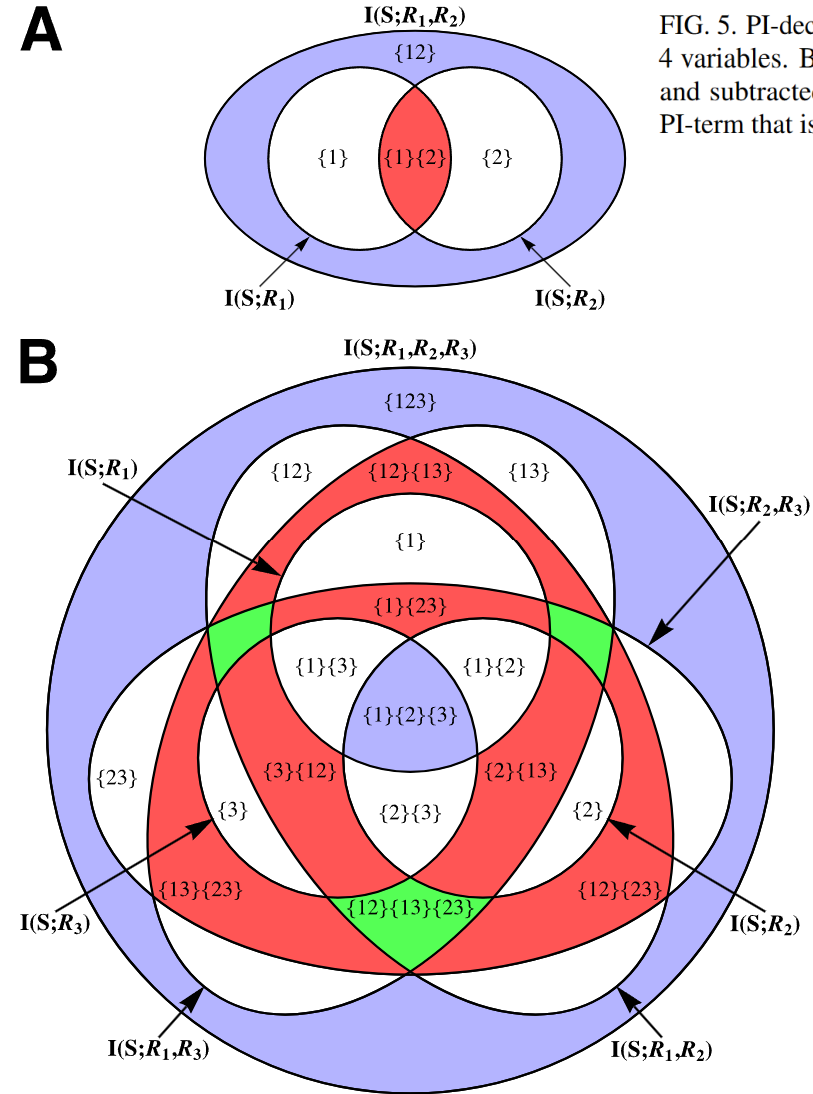
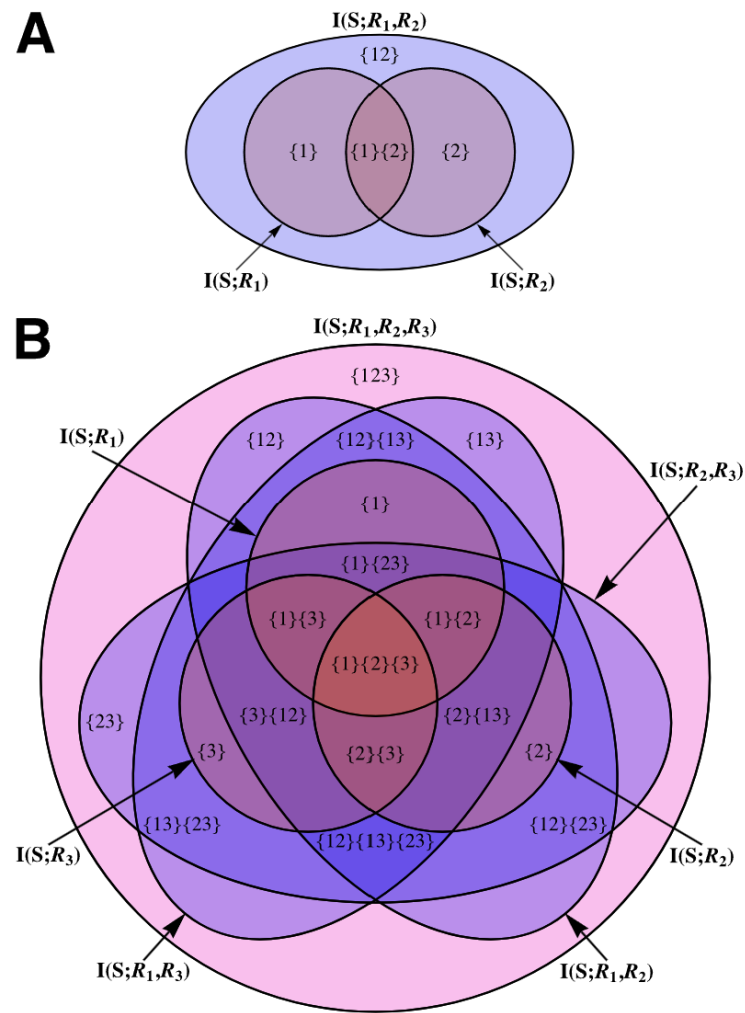


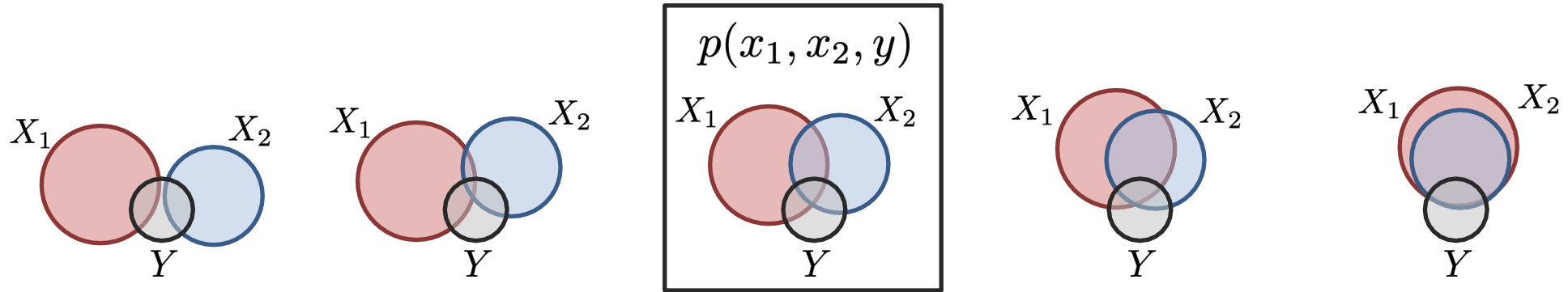
FIG. 5. PI-decomposition of interaction information for (A) 3 and (B) 4 variables. Blue and red regions represent PI-terms that are added and subtracted, respectively. The green region in (B) represents a PI-term that is subtracted twice.

Partial Information Decomposition

One type of information decomposition

Unimodal marginal-matching distributions:

$$\Delta_p = \{q(x_1, x_2, y) : q(x_1, y) = p(x_1, y), q(x_2, y) = p(x_2, y)\}$$



$$S = \underbrace{I_p(X_1, X_2; Y)}_{\text{Task-relevant multimodal info}} - \underbrace{\min_{q \in \Delta_p} I_q(X_1, X_2; Y)}_{\text{Task-relevant multimodal info without synergy:}}$$

Task-relevant multimodal info

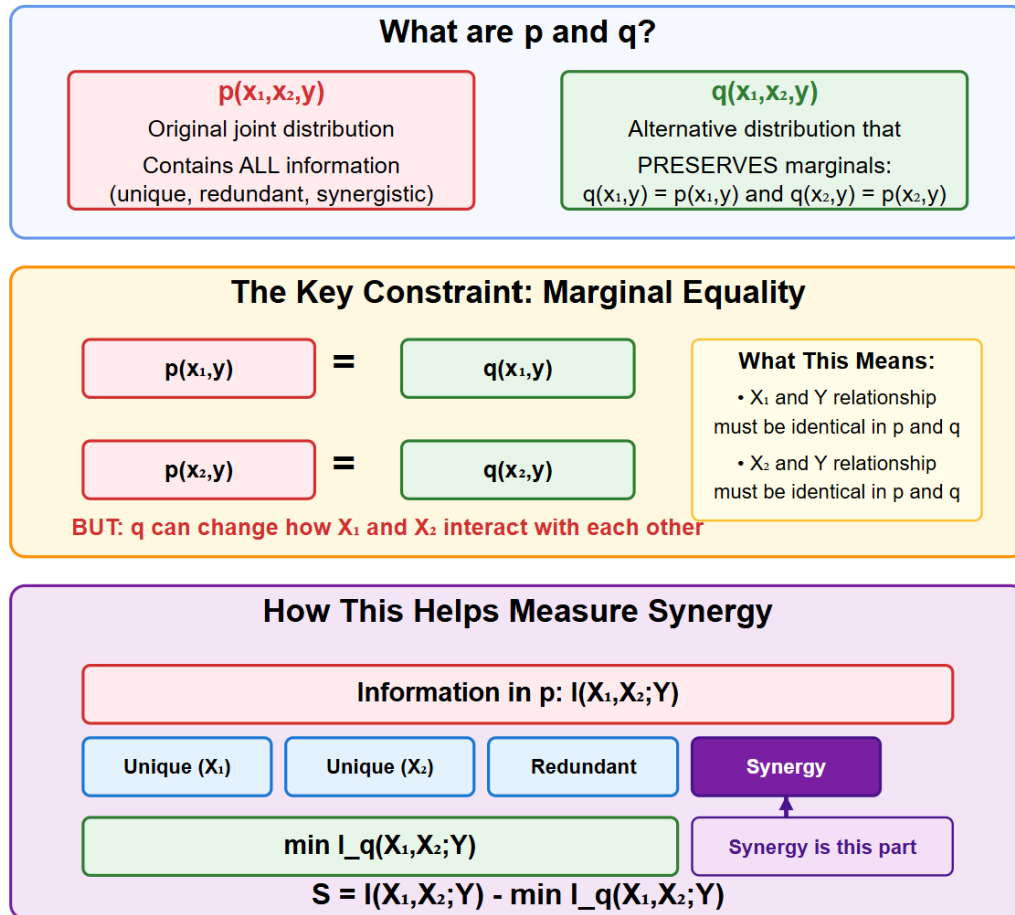
Task-relevant multimodal info without synergy:

$$S_{q^*} = I_{q^*}(X_1, X_2; Y) - \min_{q \in \Delta_p} I_q(X_1, X_2; Y) = 0$$

Partial Information Decomposition

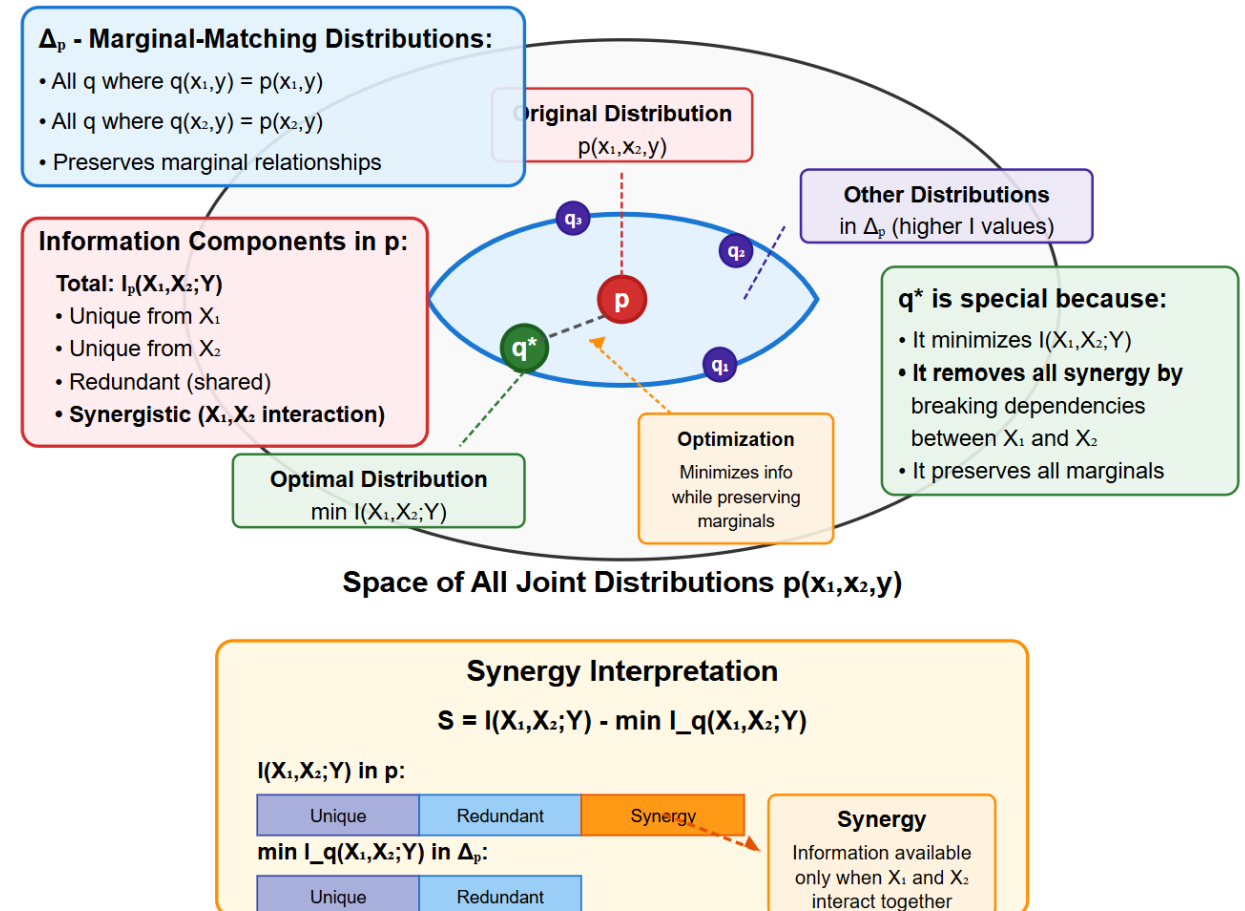
Understanding p and q in Partial Information Decomposition

How q helps isolate synergy in Bertschinger's approach



Geometric Representation of p and q

Bertschinger et al.'s Approach to Quantifying Unique Information



Partial Information Decomposition

One type of information decomposition

Unimodal marginal-matching distributions:

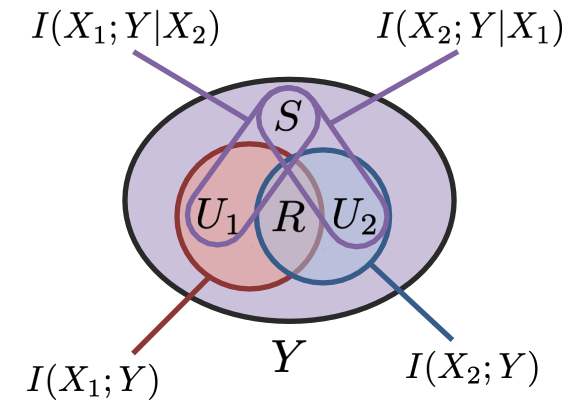
$$\Delta_p = \{q(x_1, x_2, y) : q(x_1, y) = p(x_1, y), q(x_2, y) = p(x_2, y)\}$$

$$S = I_p(X_1, X_2; Y) - \min_{q \in \Delta_p} I_q(X_1, X_2; Y)$$

+ consistency equations relating interactions with information theory:

Only need unimodal marginals to infer redundancy and uniqueness:

$$R = \max_{q \in \Delta_p} I_q(X_1; X_2; Y) \quad U_1 = \min_{q \in \Delta_p} I_q(X_1; Y|X_2) \quad U_2 = \min_{q \in \Delta_p} I_q(X_2; Y|X_1)$$



Can be solved efficiently as a convex optimization problem

Scales to high-dimensional continuous modalities via neural networks

Partial Information Decomposition

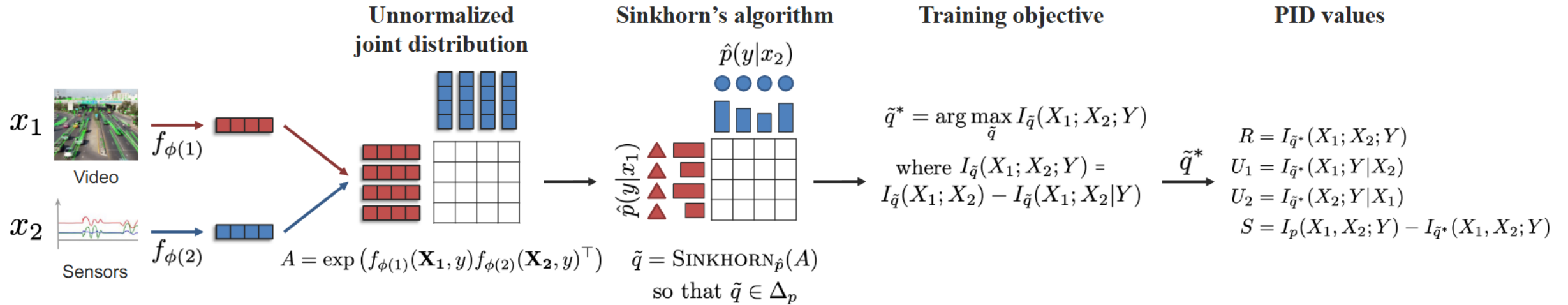


Figure 2: We propose BATCH, a scalable estimator for PID over high-dimensional continuous distributions. BATCH parameterizes $\tilde{q}$ using a matrix A learned by neural networks such that mutual information objectives over $\tilde{q}$ can be optimized via gradient-based approaches over minibatches. Marginal constraints $\tilde{q} \in \Delta_p$ are enforced through a variant of the Sinkhorn-Knopp algorithm on A .

Quantifying Interactions

Open challenges

These interactions can be efficiently estimated – gives a path towards understanding interactions

Language: *And he I don't think he got mad when hah*

I don't know maybe.

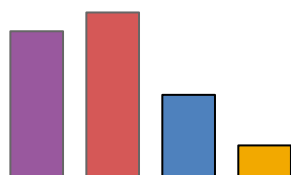
Vision:



Acoustic:

(frustrated voice)

Sentiment



$R U_\ell U_{av} S$

Language/Agreement

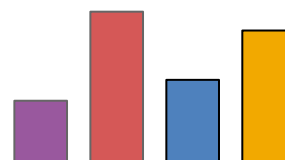
Sheldon :

Its just a *privilege* to watch your mind at work.

- **Text** : suggests a compliment.
- **Audio** : neutral tone.
- **Video** : straight face.

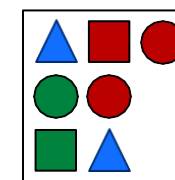


Sarcasm



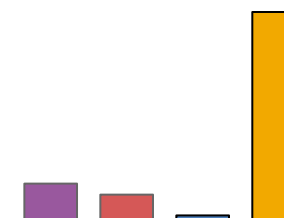
$R U_\ell U_{av} S$

Multimodal Transformer



Is there a red shape above a circle?

VQA



$R U_\ell U_i S$

Multiplicative/Transformer

Also matches human judgment of interactions, and other sanity checks on synthetic datasets

Can also be used to choose most appropriate models – can they be used to better train/design new models?

Quantifying Interactions

Open
challenges

Lower and upper bounds for interactions in a semi-supervised setting: $p(x_1, y), p(x_2, y), p(x_1, x_2)$

Idea 2: min-entropy couplings

Efficient approximation algorithms
[Cicalese et al., 2002, Compton 2022]

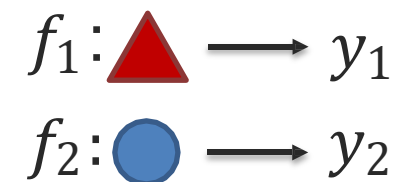
Upper bound: $\bar{S} = c_2 - \min_{r \in \Delta_p} H_r(X_1, X_2, Y) - \min_{q \in \Delta_p} I_q(X_1, X_2; Y)$

$$\bar{S} = \max_{r \in \Delta_p} I_r(X_1, X_2; Y) - \min_{q \in \Delta_p} I_q(X_1, X_2; Y)$$

$$S = I_p(X_1, X_2; Y) - \min_{q \in \Delta_p} I_q(X_1, X_2; Y)$$

Lower bound: $\underline{S} = \underbrace{\alpha(f_1, f_2) \cdot c_1}_{\text{Task-relevant multimodal info}} - \underbrace{\max(U_1, U_2)}_{\text{Task-relevant multimodal info}}$

Idea 1: disagreement

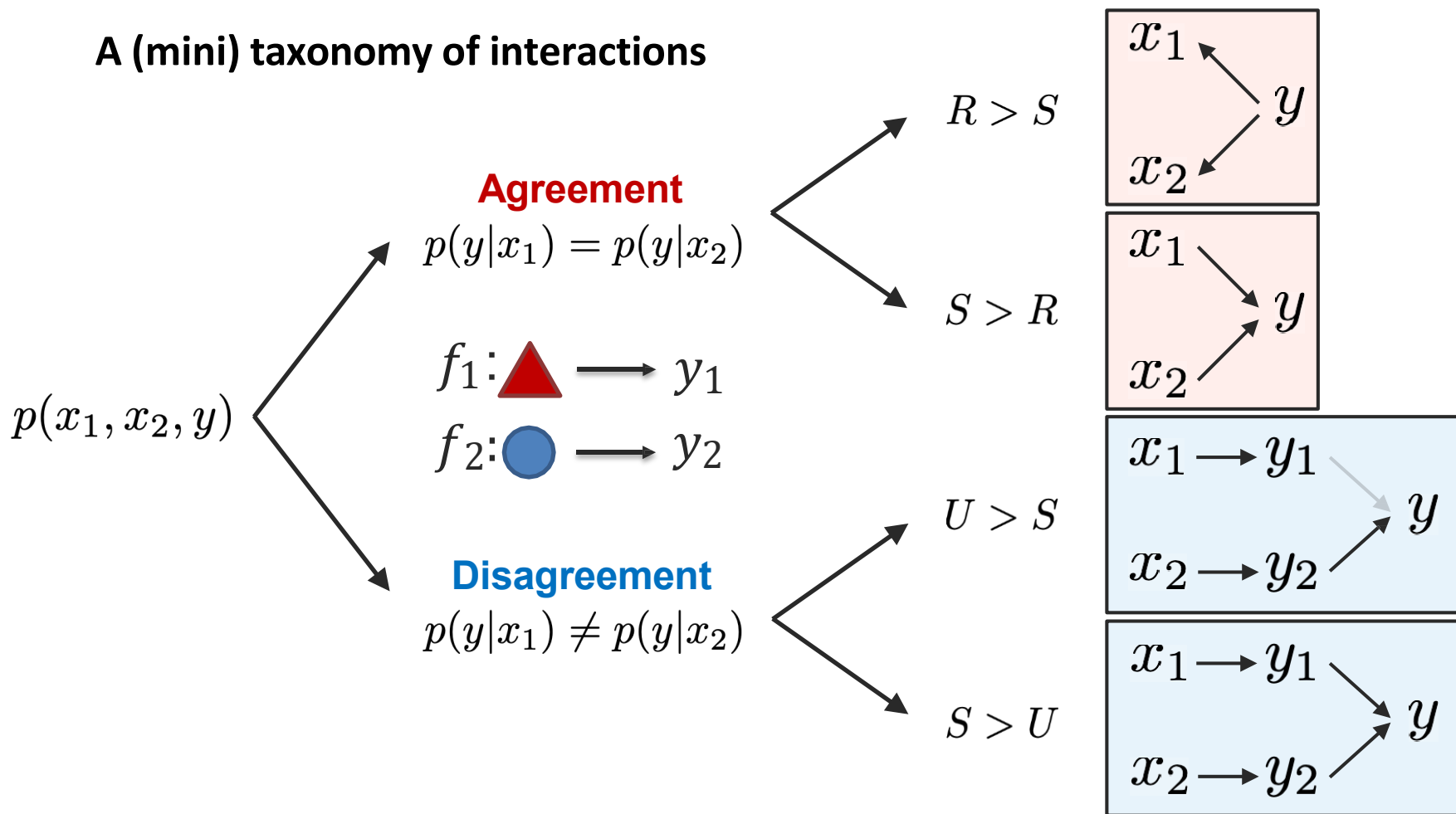


Gives theoretical results on estimating interactions and model performance for semi-supervised multimodal learning without synergy

On Agreement, Disagreement, and Synergy

Open
challenges

A (mini) taxonomy of interactions



$I(X_1; X_2) > I(X_1; X_2|Y)$
Agreement redundancy
Contrastive learning

$I(X_1; X_2) < I(X_1; X_2|Y)$
Agreement synergy
Future work?

Disagreement uniqueness
Feature selection

Disagreement synergy
Future work?

[Blum and Mitchell. Combining Labeled and Unlabeled Data with Co-training. COLT 1998]

[Peng et al., Feature selection based on mutual information criteria of max-dependency, max-relevance, and min-redundancy. TPAMI 2005]

[Liang et al., Multimodal Learning Without Labeled Multimodal Data: Guarantees and Applications, arXiv 2023]

Factorized Learning of Shared + Unique Information

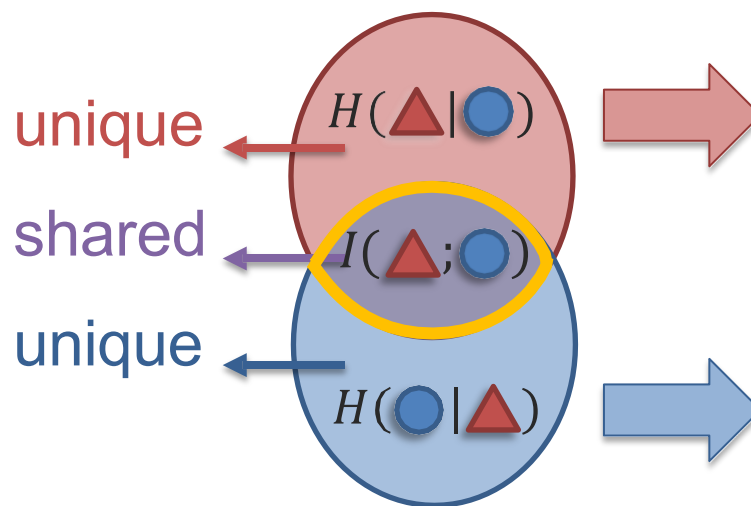
Modeling unique information



Can you please pass the cow?

Modality A 

Modality B 



- 1 Maximize the mutual information

$$I(\mathbf{z}; \bullet) \quad \text{and} \quad I(\mathbf{z}; \blacktriangle)$$

- 2 Minimize the conditional entropy

$$H(\mathbf{z}|\bullet) \quad \text{and} \quad H(\mathbf{z}|\blacktriangle)$$

Factorized Learning of Shared + Unique Information

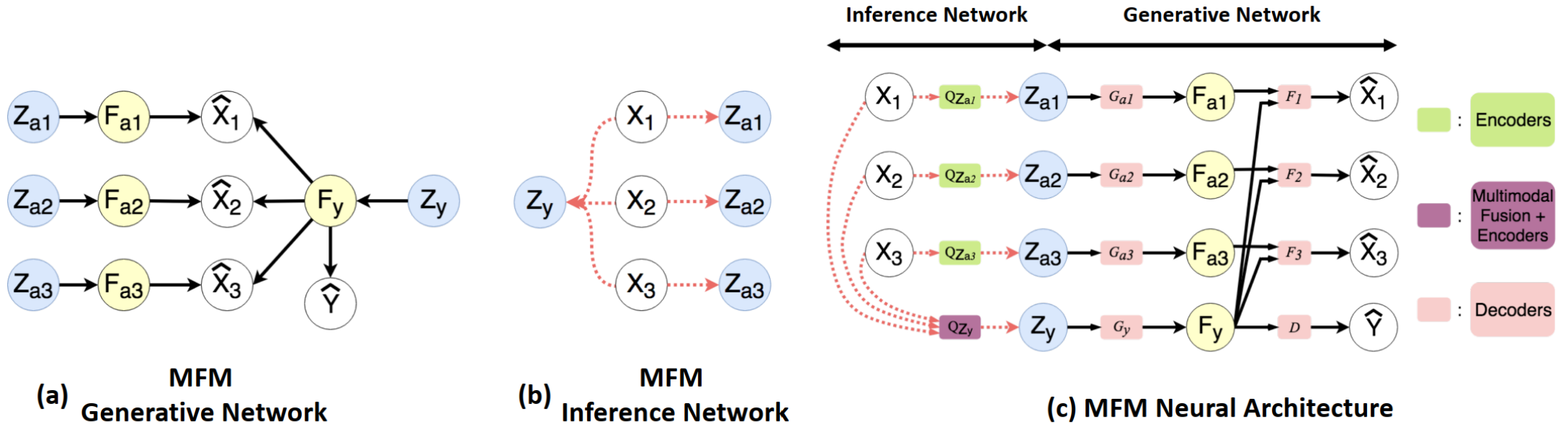


Figure 1: Illustration of the proposed Multimodal Factorization Model (MFM) with three modalities. MFM factorizes multimodal representations into *multimodal discriminative* factors $\mathbf{F}_y$ and *modality-specific generative* factors $\mathbf{F}_{\mathbf{a}\{1:M\}}$. (a) MFM Generative Network with latent variables $\{\mathbf{Z}_y, \mathbf{Z}_{\mathbf{a}\{1:M\}}\}$, factors $\{\mathbf{F}_y, \mathbf{F}_{\mathbf{a}\{1:M\}}\}$, generated multimodal data $\hat{\mathbf{X}}_{1:3}$ and labels $\hat{\mathbf{Y}}$. (b) MFM Inference Network. (c) MFM Neural Architecture. Best viewed zoomed in and in color.

Factorized Learning of Shared + Unique Information

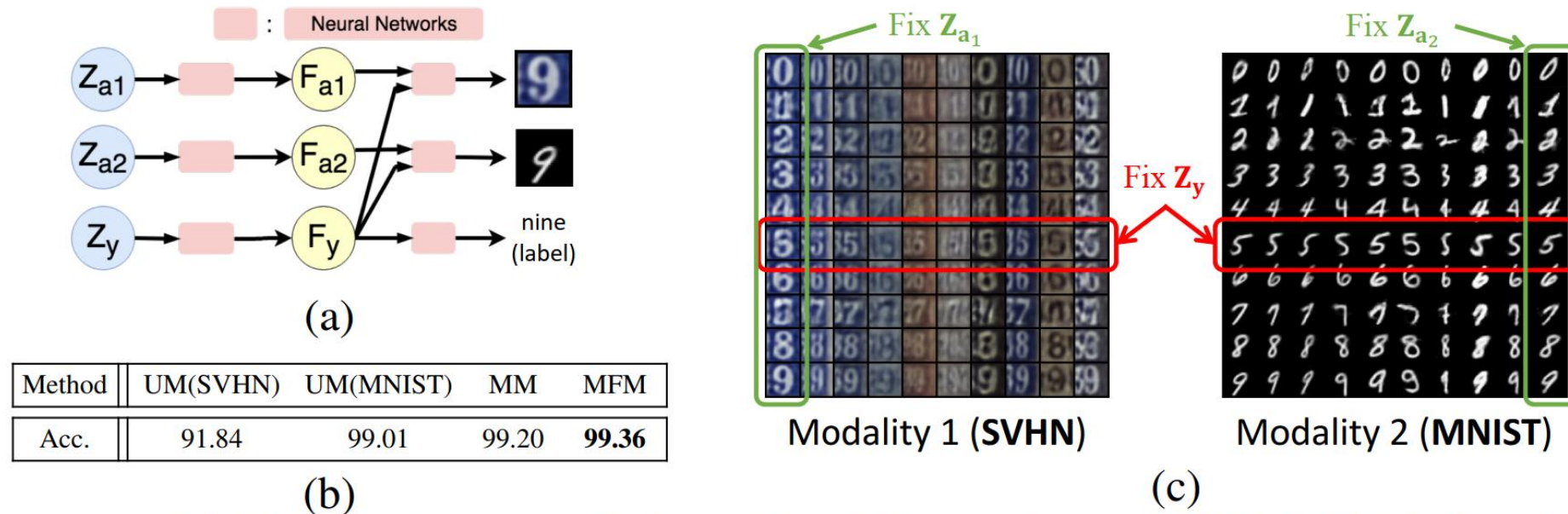
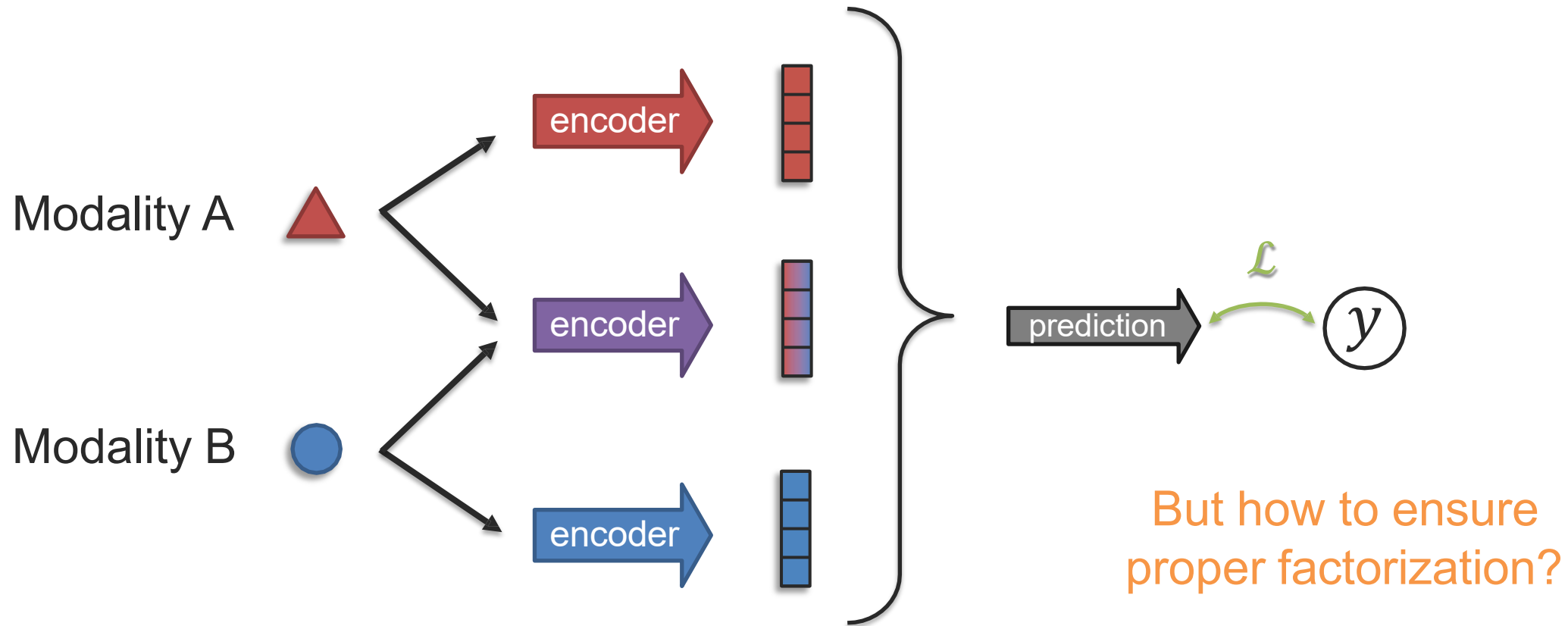
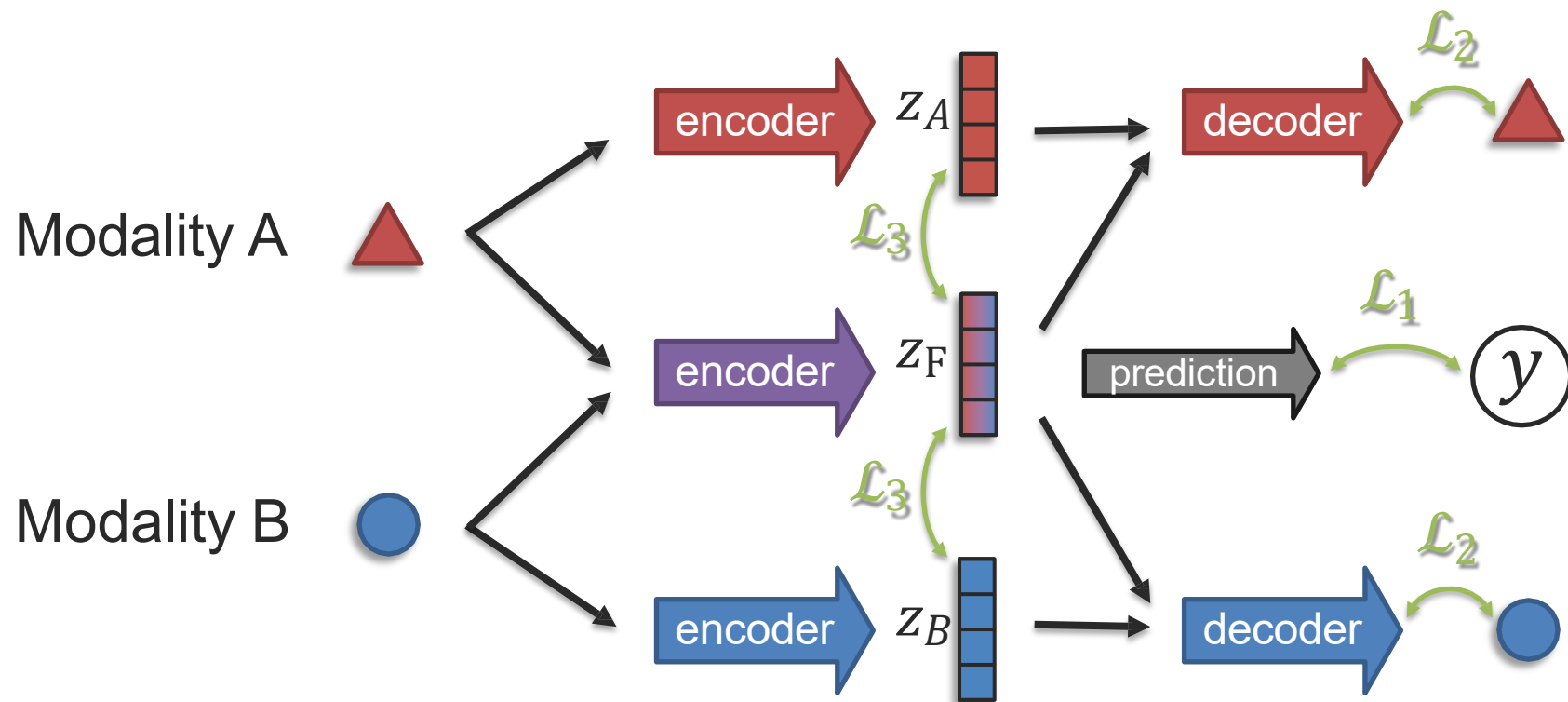


Figure 2: (a) MFM generative network for multimodal image dataset SVHN+MNIST, (b) unimodal and multimodal classification accuracies, and (c) conditional generation for SVHN and MNIST digits. MFM shows improved capabilities in digit prediction as well as flexible generation of both images based on labels and styles.

Factorized Multimodal Representations



A Generative-Discriminative Approach



$$\mathcal{L} = \mathcal{L}_1 + \mathcal{L}_2 + \mathcal{L}_3$$

$\mathcal{L}_1$: discriminative

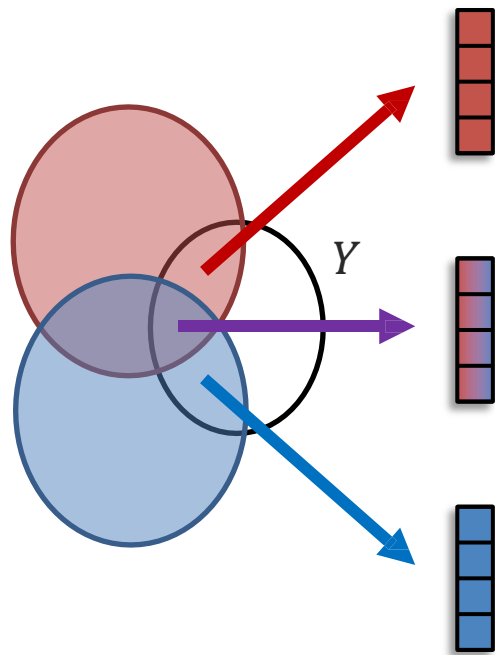
$\mathcal{L}_2$: generative

$\mathcal{L}_3$: no overlap

Independent priors
for z_A , z_B and z_F

Learning Task-relevant Unique Information

Modeling task-relevant unique information



- 2 Maximize task-relevant **unique** information

$$I(\mathbf{Z}; Y | \bullet)$$

- 1 Maximize task-relevant **shared** information

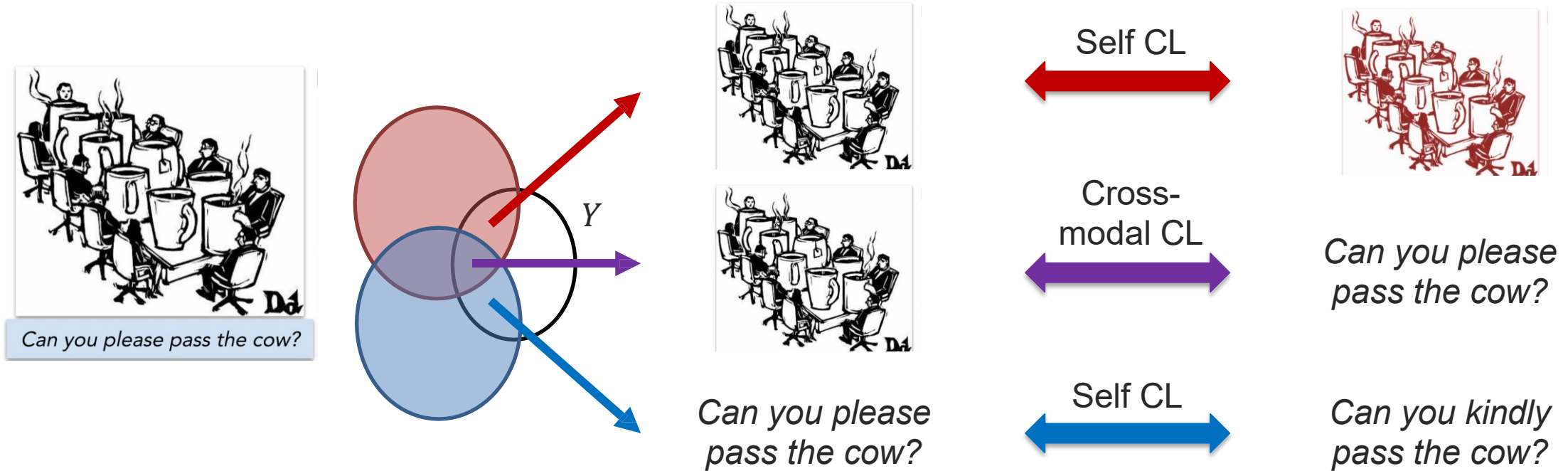
$$I(\mathbf{Z}; \bullet; Y) \quad \text{and} \quad I(\mathbf{Z}; \blacktriangle; Y)$$

- 3 Maximize task-relevant **unique** information

$$I(\mathbf{Z}; Y | \blacktriangle)$$

Learning Task-relevant Unique Information

Modeling task-relevant unique information



Approximate task-relevance Y using multi-view data augmentations
New scalable lower and upper bounds on mutual information

Learning Task-relevant Unique Information

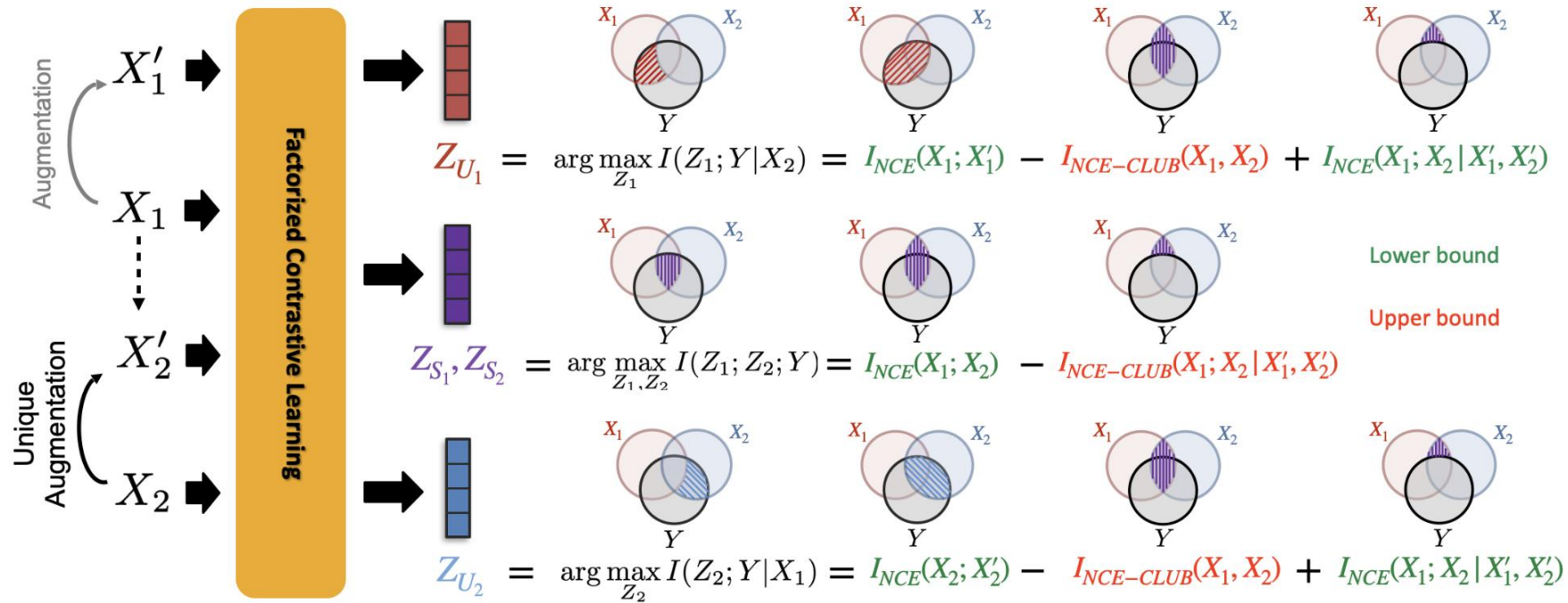
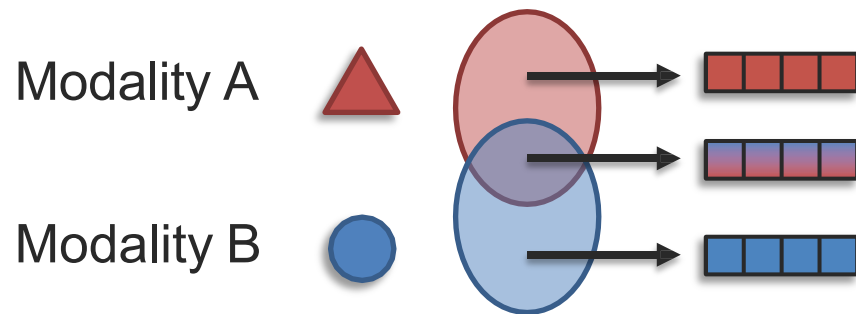


Figure 2: FACTORCL: We propose a self-supervised CL method to learn *factorized* representations Z_{S_1} , Z_{S_2} , Z_{U_1} , and Z_{U_2} to capture task-relevant information shared in both X_1 and X_2 , unique to X_1 , and unique to X_2 . By starting with information-theoretic first principles of shared and unique information, we design contrastive estimators to both *capture task-relevant* and *remove task-irrelevant* information, where a notion of task-relevance without explicit labels is afforded by a new definition of *multimodal augmentations* X'_1, X'_2 . Lower bounds are in green and upper bounds are in red.

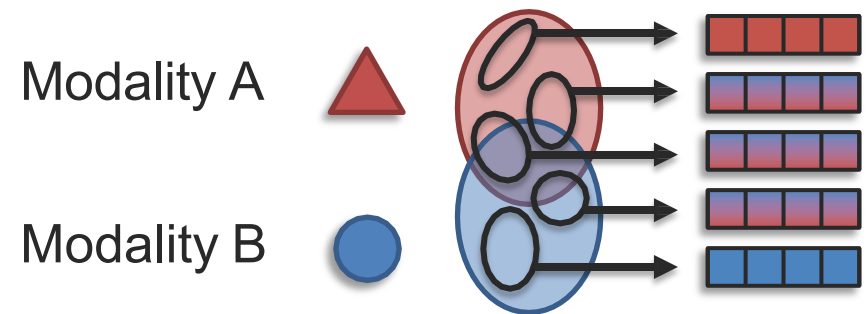
Fine-Grained Fission

How to automatically discover these internal clusters, factors?

Modality-level fission:



Fine-grained fission:

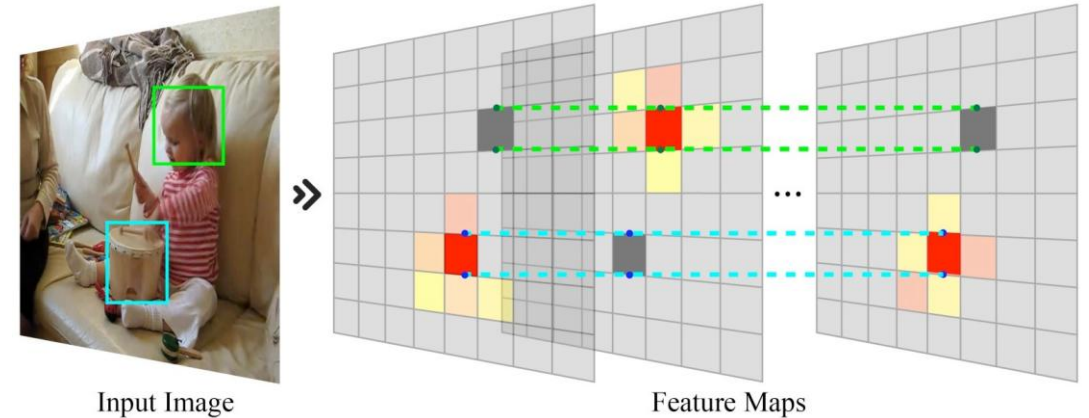
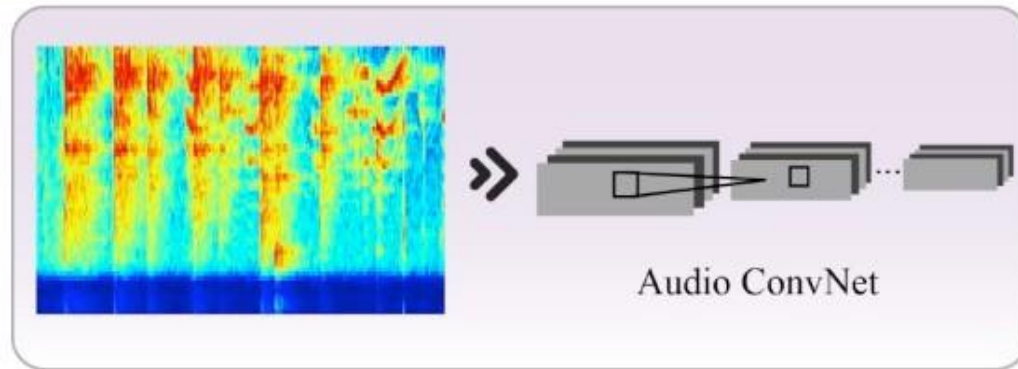


Fine-Grained Fission – A Clustering Approach

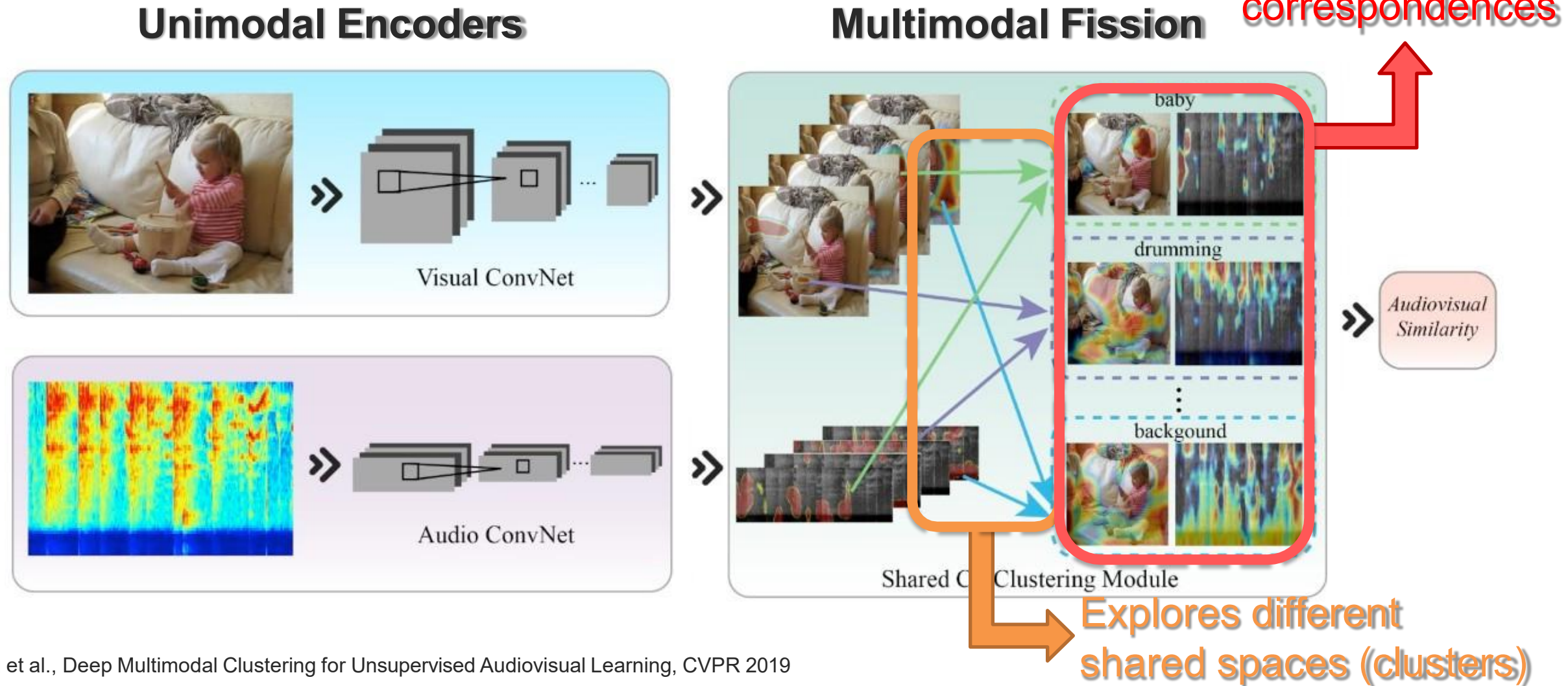
Unimodal Encoders



Localized activations for different objects



Fine-Grained Fission – A Clustering Approach

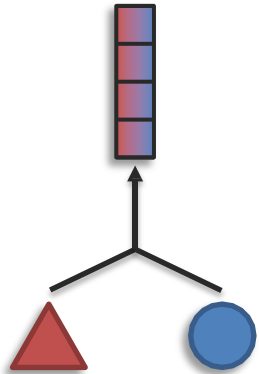


Challenge 1: Representation

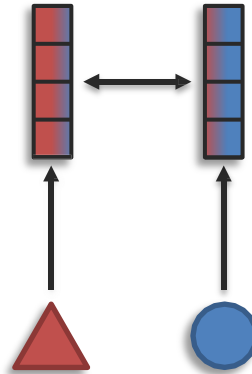
Definition: Learning representations that reflect cross-modal interactions between individual elements, across different modalities

Sub-challenges:

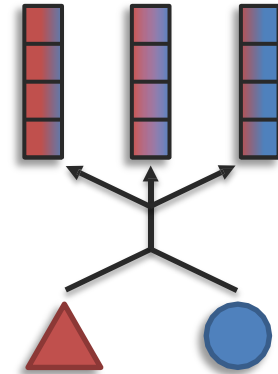
Fusion



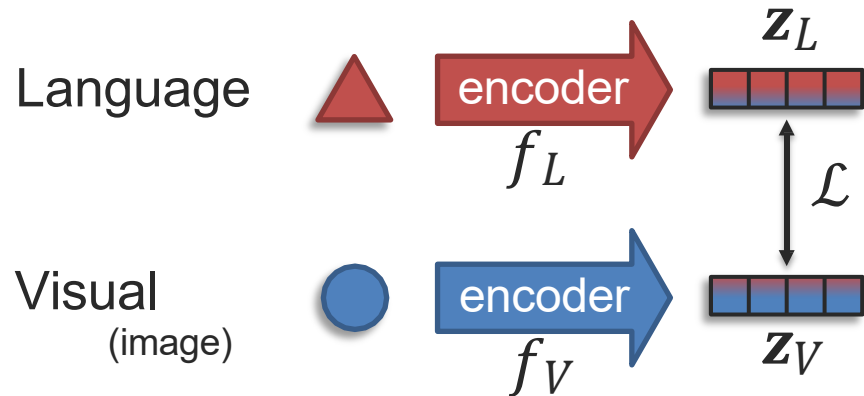
Coordination



Fission



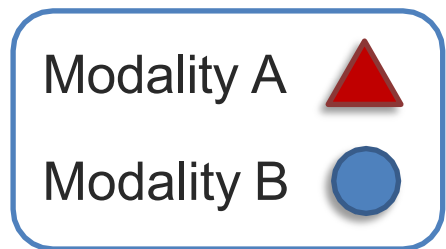
Recap: Contrastive Learning and Connected Modalities



Popular contrastive loss: InfoNCE

$$\mathcal{L} = -\frac{1}{N} \sum_{i=1}^N \log \frac{\text{sim}(\mathbf{z}_A^i, \mathbf{z}_B^i)}{\sum_{j=1}^N \text{sim}(\mathbf{z}_A^i, \mathbf{z}_B^j)}$$

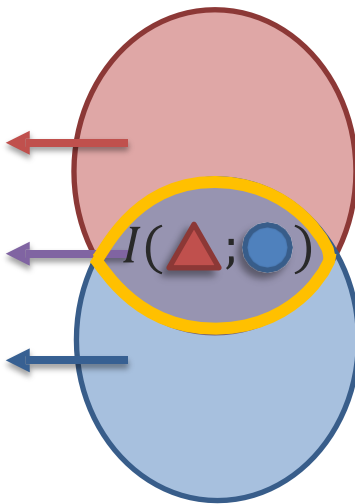
Connected modalities:



unique

shared

unique



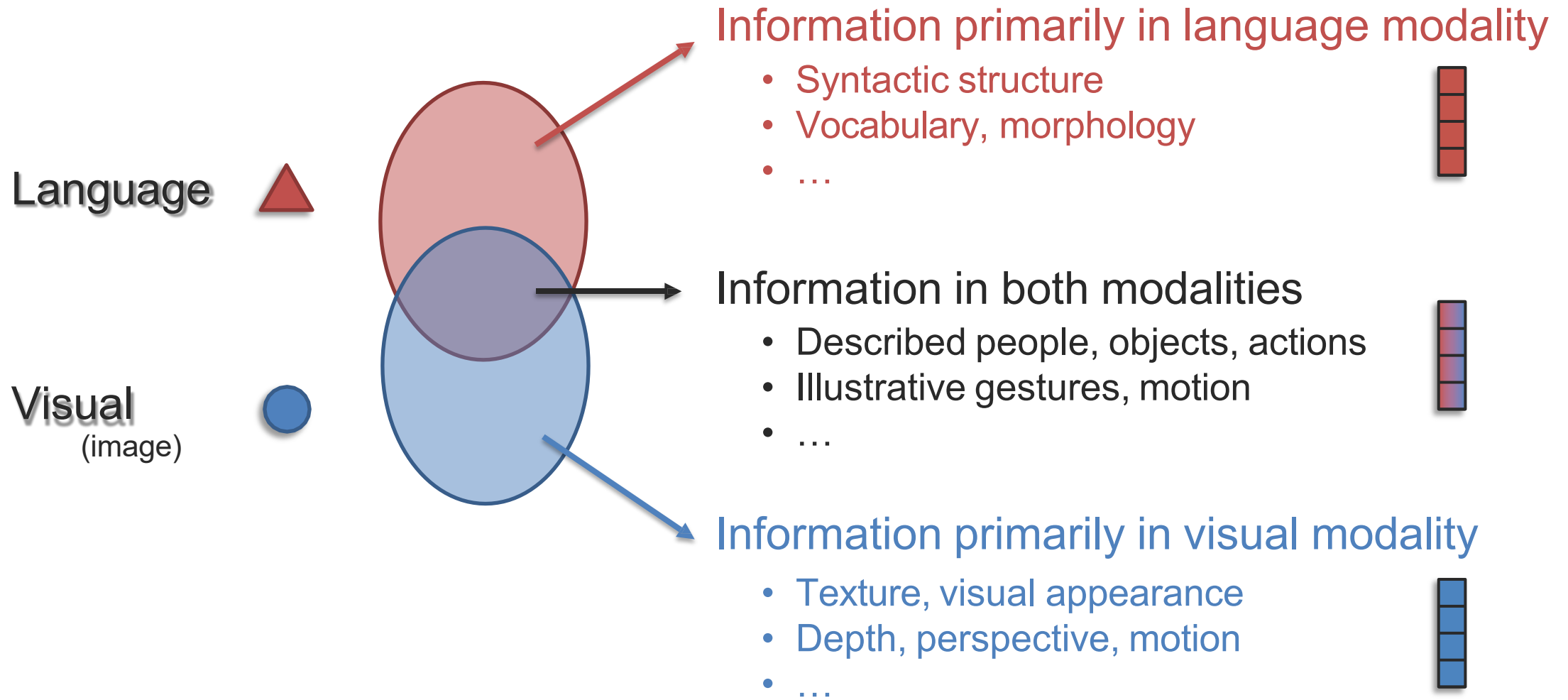
Mutual information $I(X; Y)$

$$D_{X,F} \left[\log \frac{P_{XF}(x, y)}{P_X(x)P_F(y)} \right]$$



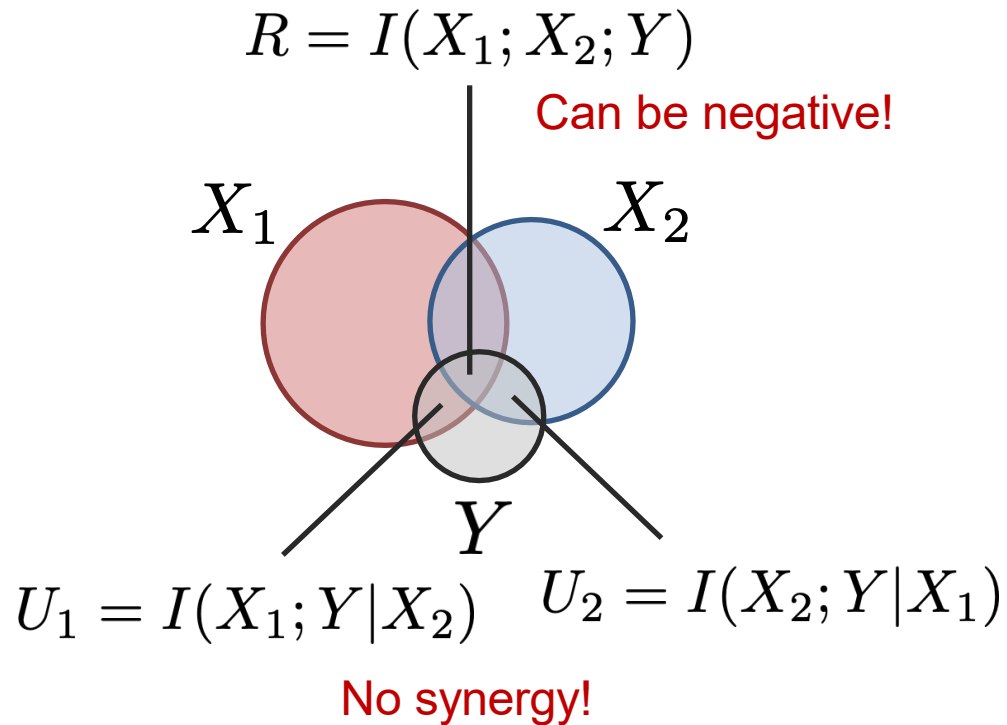
CLIP focuses on shared connections

Recap: Modality-Level Fission

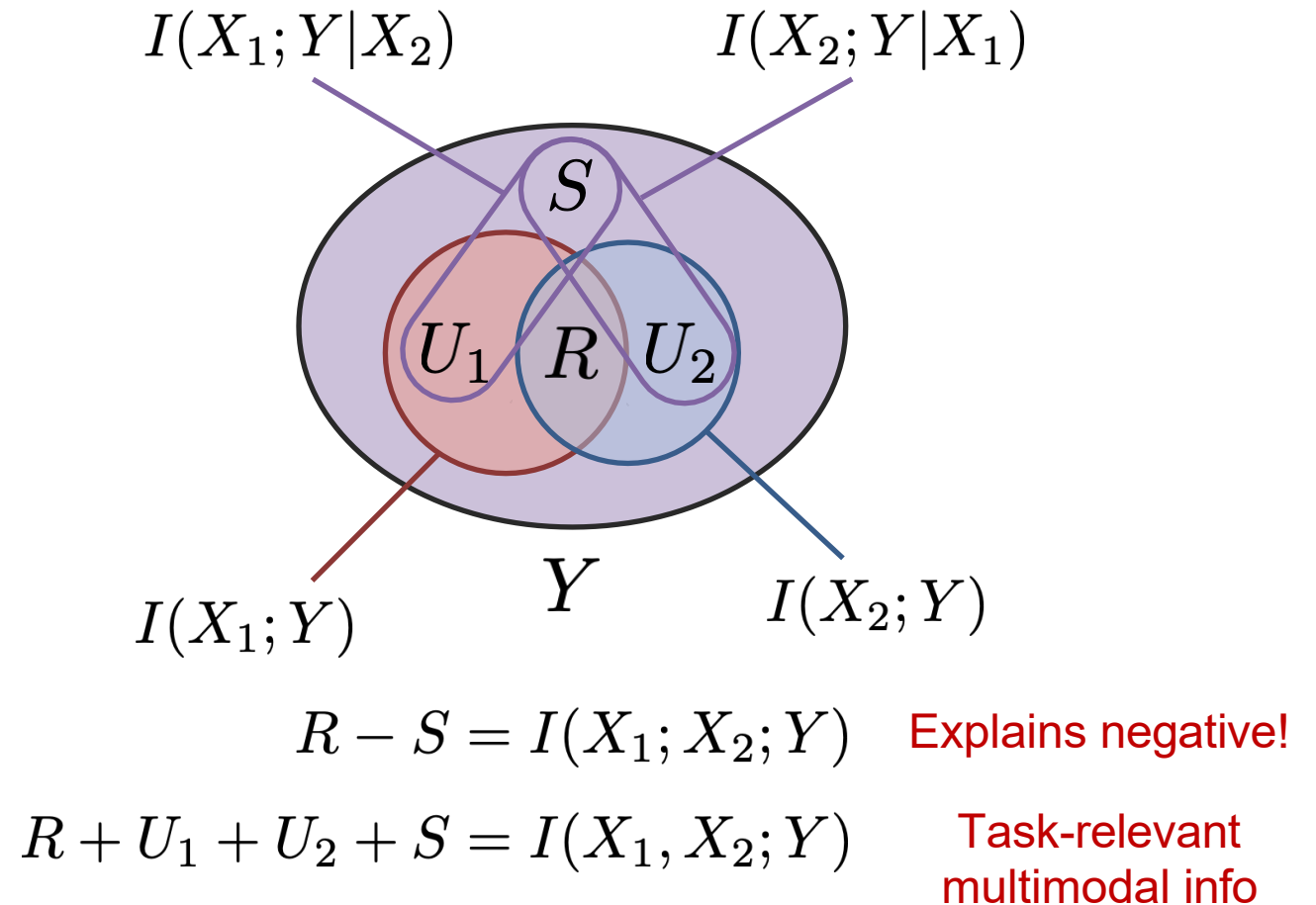


Recap: Partial Information Decomposition

Classical Information Theory



Partial Information Decomposition



References

- Multi Modal Machine Learning, 11-777 • Fall 2023 • Carnegie Mellon University